

Night-Time Public Transport Use in London: Temporal Typologies and Socio-Spatial Contexts of Rail and Bus Activity

Fangzheng Zhou

CASA0010: Dissertation 25/26

Supervisor: Esra Suel, Clara Peiret-García

Word Count: 11205

Code Repository:

<https://github.com/fangzhengz/>

Night-Time-Public-Transport-Use-Dissertation

This dissertation is submitted in part requirement for the MSc in the
Centre for Advanced Spatial Analysis, Bartlett Faculty of the Built Environment, UCL

Submission Date: 20 August 2026

Abstract

Urban transport planning has traditionally prioritised daytime commuting, leaving night-time mobility comparatively less understood despite the growing importance of London's 24-hour economy. This dissertation examines how public transport use between 18:00 and 05:00 is organised across London's Rail and Bus systems, how the two modes differ, and how these patterns relate to night-work geography and wider socio-spatial context.

Using Transport for London's NUMBAT 2024 and BUSTO 2024/25 datasets, temporal profiles were constructed for 403 Rail stations and 3,383 bus-served LSOAs. Gaussian Mixture Models, applied separately to each mode, identified recurring night-time usage patterns. These were characterised by their temporal, directional, day-type and spatial properties, then related to the London Night Workers Classification (LNWC) and 20 area-level socio-economic, housing and employment indicators.

The results show that night-time public transport use is not a homogeneous off-peak condition. Rail activity is differentiated by direction, spatial position, interchange role and late-night duration: central stations are generally more departure-oriented and outer stations more arrival-oriented, but persistence follows a different pattern, remaining highest at some lower-volume inner and eastern stations rather than the most intensive central night-work areas. Bus use instead follows a smoother gradient from persistent central and inner activity towards earlier-declining outer areas. Night-work geography corresponds most clearly with these central-peripheral contrasts but does not fully account for the observed rhythms. These central clusters are further associated with younger populations, private renting, low car access and high facility intensity, while the outer clusters combine weaker realised activity with relatively high resident employment in night-related sectors, suggesting a possible divergence between the residential geography of night-related employment and the concentration of night-time transport activity.

The study demonstrates the value of treating the night as a dedicated analytical window and analysing Rail and Bus separately, providing a reproducible framework for relating transport-derived patterns to independently measured urban context while distinguishing realised transport use from underlying demand.

Declaration

I hereby declare that this dissertation is all my own original work and that all sources have been acknowledged.

It is 11,205 words in length.

Signed: Fangzheng Zhou

Date: 20 August 2026

Contents

Abstract	2
Declaration	3
List of Acronyms and Abbreviations	8
Acknowledgements	9
1 Introduction	10
1.1 Research Background	10
1.2 Research Question	11
1.3 Dissertation Structure	12
2 Literature Review	14
2.1 The Urban Night and Night-time Activity	14
2.2 Night-time Transportation and Mobility	15
2.3 Smart-card Data, Temporal Profiles and Socio-spatial Interpretation	16
2.4 Synthesis: Research Gap and Thesis Positioning	18
3 Methodology	20
3.1 Research Design and Analytical Framework	20
3.2 Data and Preparation	21
3.2.1 Rail Passenger Activity: NUMBAT 2024	21
3.2.2 Bus Passenger Activity: BUSTO 2024/25	22
3.2.3 Urban Night Contextual Data and Preparation	24
3.2.4 Socio-economic, Demographic and Functional Indicators	26
3.3 Analytical Methods	30
3.3.1 Feature Engineering	30
3.3.2 Gaussian Mixture Model Clustering	34
3.3.3 Post-clustering Behavioural Descriptors	36
3.3.4 Post-clustering Association with LNWC and Urban Background Indicators	38
3.4 Ethical Considerations and Methodological Boundaries	43
4 Results	44
4.1 Selection of the Cluster Solutions	44
4.2 Night-time Public Transport Usage Types	47
4.2.1 Five Night-time Usage Types: Rail System	47
4.2.2 Four Night-time Area Profiles: Bus System	53
4.3 Relationship with Area Context	57
4.3.1 Correspondence with the Geography of Night Work	58
4.3.2 Relation to City Background Variables	60
4.4 Summary of Results	67

5	Discussion	68
5.1	Overall Organisation of Night-time Public Transport Use	68
5.2	Different Forms of Night-time Organisation in Rail and Bus	69
5.3	Systematic but Incomplete Correspondence with Night-work and Wider Area Context	70
5.4	Exceptional Types and Night-time Environments Requiring Closer Attention . .	71
5.5	Implications for Night-time Transport Research and Planning	72
5.6	Limitations and Future Research	73
6	Conclusion	74
	Bibliography	76
	Appendices	80
A	Supplementary Figures and Tables	80
B	Research Log and Research Reflection	94
B.1	Research Log	94
B.2	Research Process Reflection	102

List of Figures

3.1	Research framework	21
3.2	LNWC spatial distribution	26
3.3	Rail station catchment zones	40
4.1	Rail model-selection diagnostics	45
4.2	Bus model-selection diagnostics	46
4.3	Rail station temporal profiles	48
4.4	Rail cluster map	49
4.5	Rail behavioural descriptors	50
4.6	Bus temporal profiles	54
4.7	Bus cluster map	55
4.8	Bus behavioural descriptors	56
4.9	Rail LNWC enrichment	58
4.10	Bus LNWC enrichment	60
4.11	Rail contextual effect-size ranking	62
4.12	Rail contextual profiles	63
4.13	Bus contextual effect-size ranking	65
4.14	Bus contextual profiles	66
A.1	Rail usage types and LNWC context	80
A.2	Bus usage types and LNWC context	81
A.3	Behavioural-descriptor effect sizes	81
A.4	Rail time-series feature map	82
A.5	Bus time-series feature map	83

List of Tables

3.1	LNWC cluster types	25
3.2	Contextual variables	27
3.3	Rail and Bus feature matrices	33
3.4	Post-clustering behavioural descriptors	36
4.1	Strongest contextual differences	61
A.1	Rail behavioural descriptors	84
A.2	Bus behavioural descriptors	85
A.3	Contextual-indicator omnibus tests	86
A.4	Rail cluster-versus-rest contextual effects	90
A.5	Bus cluster-versus-rest contextual effects	92
B.1	Research log	95

List of Acronyms and Abbreviations

ARI Adjusted Rand Index

BH Benjamini–Hochberg

BIC Bayesian Information Criterion

CLR Centred Log-Ratio

GMM Gaussian Mixture Model

IMD Indices of Multiple Deprivation

LNWC London Night Workers Classification

LSOA Lower Layer Super Output Area

NaPTAN National Public Transport Access Nodes

ONS Office for National Statistics

POI Point of Interest

SIC Standard Industrial Classification

TfL Transport for London

Acknowledgements

First and foremost, I would like to express my sincere gratitude to my supervisors, Dr. Esra Suel and Clara Peiret-García, for their invaluable guidance and support throughout my research. Their expertise and insights have been instrumental in shaping this study. Additionally, I extend my heartfelt thanks to partner supervisors from TfL, Howard Wong and Mikaella Mavrogeni. Your patient guidance and professional knowledge have also played a crucial role in this research, especially in terms of data and analysis methods.

I am also deeply grateful to my classmates, friends, and family for their unwavering support and encouragement during this challenging research process. Their belief in me has been a constant source of motivation.

Lastly, I extend my best wishes to my advisors, friends, and myself for success and fulfillment in our future academic and research endeavors.

Chapter 1

Introduction

1.1 Research Background

In recent years, the “24-hour city” has become an increasingly important concept in urban policy. London is a prominent example: the introduction of the Night Tube in 2016, together with subsequent policies on the night-time economy, culture and governance (Transport for London, 2016; GLA Economics, 2024), signals that city authorities no longer treat the night simply as a pause in daytime activity, but increasingly recognise it as a period with independent social and economic value (Shaw, 2010, 2014).

Against this backdrop, around one quarter of London’s workforce is estimated to work during night-time hours (Mavrogeni et al., 2025), and the mobility demands generated by night-time employment, consumption and leisure have become an important challenge for the transport system.

However, urban transport planning has long been organised around daytime activity and commuting, shaping service design, infrastructure allocation and the evaluation frameworks used to assess transport systems (McArthur et al., 2019). Academic research on how cities function after dark has also remained relatively limited (Van Liempt et al., 2015). This daytime orientation in both planning and research is particularly problematic because nighttime travel is not simply a

temporal extension of daytime travel. The transport environment at night differs: public transport services are typically more limited, lighting and safety conditions change, and travellers' purposes and resources are more diverse (Kapitza, 2022, 2025). Night-time transport must serve not only leisure and social activities, but also the mobility of night-shift workers and other time-constrained groups (Plyushteva, 2019; Shaw, 2022; Palm et al., 2023). These characteristics create particular challenges for understanding when and where night-time transport demand occurs.

High-temporal-resolution smart-card data and clustering methods have been widely used to identify typical usage patterns in public transport systems (Briand et al., 2016; El Mahrsi et al., 2017; Manley et al., 2018; Gan et al., 2020; Zhao et al., 2020). However, these analyses have typically focused on weekday commuting, peak-hour flows or all-day ridership structures. The night-time period has more often appeared as part of the full daily cycle than as a distinct analytical window.

Thus, it remains unclear whether stable and recurring temporal usage patterns exist within night-time public transport, and how these patterns are distributed across urban space.

This study addresses this gap by analysing high-temporal-resolution smart-card data for London's rail and bus systems. It identifies temporal patterns of night-time public transport use, examines their spatial distribution, and situates them against the geography of London's night-time workforce (Mavrogeni et al., 2025) and broader socio-spatial context.

1.2 Research Question

Building on this background, the study poses the following research question:

Using high-temporal-resolution public transport usage data, what distinct patterns of night-time public transport usage can we observe within London's rail and bus systems, and what are their temporal and spatial characteristics? What differences exist in the night-time usage patterns of these two modes of transport, and how do they relate to the geographical distribution of

night-time work in London and the broader socio-spatial context at the area level?

This question is addressed through two specific sub-questions:

RQ1: What distinct night-time usage types can be identified for rail stations and bus-served LSOAs, based on their temporal, directional and day type activity patterns? How are these types distributed across London and in what respects do the two transport modes differ?

RQ2: To what extent are the identified usage types associated with the London Night Workers Classification (LNWC) and with local-level socio-economic, household, housing and employment contexts? How do these associations help us understand the urban environments under the different usage types?

To address these questions, the study first identifies night-time usage types separately within the Rail and Bus systems, then characterises their temporal, directional, day-type and spatial patterns, and finally examines how the fixed usage types are associated with LNWC and wider area-level urban contexts. This sequence moves from the identification of observed transport-use patterns to their external contextual interpretation.

This study contributes by relating night-time usage types derived from high-temporal-resolution transport data to an independently constructed geography of night-time work and the broader socio-spatial context. This provides both a methodological link between temporal transport analysis and area-based contextual data, and an empirical account of how night-time transport patterns are situated across London.

1.3 Dissertation Structure

The remainder of the dissertation is organised as follows. Chapter 2 reviews literature on the urban night, night-time mobility and smart-card clustering methods. Chapter 3 describes the study area, data, feature construction, clustering methodology, and contextual analysis. Chapter 4 presents the identified rail and bus usage types, their temporal and spatial characteristics, and

their associations with the LNWC and wider socio-spatial conditions. Chapter 5 interprets these findings in relation to the research questions and wider literature, considers their implications for night-time transport research and planning, and reflects on the main methodological limitations. Chapter 6 concludes the dissertation by synthesising the answers to the research questions and summarising the study's main contribution.

Chapter 2

Literature Review

2.1 The Urban Night and Night-time Activity

The night has long been treated in urban research as an appendage of daytime activity, with its own temporal and spatial logic receiving relatively little independent attention (Van Liempt et al., 2015). Yet the urban night is not simply a continuation of the day, nor a transitional phase in which urban functions gradually wind down.

Employment, leisure, socialising, care, maintenance and return journeys all continue after conventional working hours, but at different times, in different places and under different conditions.

These differences produce patterns of urban life that are distinct from daytime commuting rhythms (Schwanen et al., 2012; Van Liempt et al., 2015).

Research that treats the night as a distinct urban period has developed largely around the concept of the “night-time economy” (NTE). Early work focused particularly on catering, bars, cultural consumption and entertainment, highlighting the contribution of night-time activity to urban economic development and cultural vitality.

However, viewing the night primarily through leisure and consumption can overlook its wider social dimensions. Shaw (2010, 2014) and Hadfield (2015) show that the urban night also involves labour, governance, regulation, safety, exclusion and inequality, while different groups experience unequal conditions of access to and participation in night-time urban space (Plyushteva, 2019). The urban night is thus not a homogeneous period defined solely by nightlife and leisure, but a setting in which different activities, populations and social conditions coexist.

This diversity is also reflected in the spatial organisation of night-time activity. Compared with the relatively stable rhythms that daytime activity forms around employment centres, educational facilities and major commuting corridors, night-time activities may be distributed across central entertainment districts, transport hubs, residential neighbourhoods, healthcare facilities, logistics areas and concentrations of night-time employment. Opening hours, public service availability, working arrangements and spatial regulation together shape when and where these activities occur. The same location may serve different urban functions by day and by night, while different parts of the city may exhibit different patterns of activity timing, duration and directional flow (Schwanen et al., 2012; Kim, 2020; Shaw, 2022).

2.2 Night-time Transportation and Mobility

The diversity of night-time activities is accompanied by a different set of travel conditions after dark. Different groups use the same transport system at different times and for different purposes (Shaw, 2022; Palm et al., 2023). Understanding night-time mobility requires attention both to the conditions under which travel takes place and to the temporal relationship between activities and transport services.

First, the operating conditions of night-time travel differ from those during the day. Using a discrete choice model, Kapitza (2022) finds that lighting, service availability and perceived safety influence mode choice at night. Other studies similarly show that personal safety can affect travellers' willingness to use public transport. Research in Brussels and Recife (Farina et al., 2022), together with analysis of the night-bus network in Sofia (Plyushteva and Boussauw, 2020),

demonstrates that night-time mobility depends not only on the existence of transport services but also on the conditions under which those services can be used. Together, these factors influence both mode choice and the practical usability of public transport at night.

Another challenge for night-time mobility is the mismatch between the temporal structure of urban activity and transport provision. Operating hours, service frequency, interchange conditions and information availability affect whether destinations can be reached at the required time. Two locations may be well connected within the transport network, but that connection becomes less useful if the required service has been reduced or withdrawn by the time it is needed (McArthur et al., 2019; Ryan et al., 2023; Pan et al., 2025). This temporal mismatch affects a wide range of night-time travellers. Research on non-standard working hours shows that altered schedules can expose workers to transport conditions different from those experienced during the day (Plyushteva, 2019; Lim et al., 2024; Kapitza, 2025), but similar constraints apply to anyone whose activity extends beyond the hours of full-service provision, including those returning from leisure, social or care activities.

This complexity is also compounded by the fact that night-time activity can be difficult to identify within datasets dominated by daytime patterns. Kim (2020) demonstrates this point using functional principal component analysis on hourly pedestrian traffic in Seoul: night-time vitality patterns emerge only when extracted as a separate component from the overwhelming daytime trend, and their temporal boundaries vary across districts according to local urban functions. The implication for transport research is that night-time usage patterns are unlikely to be adequately captured as a residual component of all-day analysis and instead require dedicated examination.

2.3 Smart-card Data, Temporal Profiles and Socio-spatial Interpretation

Studying these variations requires data that capture both when and where public transport activity occurs. Smart-card and operational transport data provide large-scale, continuous records of

passenger transactions. Such records usually do not directly reveal passengers' travel purposes or their wider activity contexts. However, their scale, continuity and temporal resolution make it possible to examine how ridership varies within the day, across the week and between locations (Zhong et al., 2016; Manley et al., 2018; Cats, 2024).

Smart-card research has developed at both individual and location-based scales. Individual-level studies compare passengers' transactions across dates and time periods to identify regular or recurring travel behaviour (Zhong et al., 2016; Manley et al., 2018). Location-based studies instead aggregate ridership at stations, routes or areas and examine how patterns vary within the day and across the week (Gan et al., 2020; Zhang et al., 2021; Cheng et al., 2024). In this second approach, stations and areas provide spatial units through which differences in urban activity rhythms can be observed.

As research has moved beyond describing total ridership towards identifying recurring temporal structures, clustering and other unsupervised classification approaches have become widely used. Briand et al. (2016), for example, use a mixture model to identify latent temporal usage types in public transport data. El Mahrsi et al. (2017) combine smart-card data processing and clustering to summarise large-scale urban mobility records, while Cheng et al. (2024) use fuzzy clustering to examine within-day and within-week ridership patterns at London rail stations. Other studies have used approaches including k-means and hierarchical clustering (Zhao et al., 2020). Despite differences in method, this literature demonstrates that complex temporal ridership records can be organised into a smaller number of interpretable profiles that share similar patterns of use.

Temporal profiling can also support the interpretation of transport patterns in relation to urban structure. Existing studies have linked ridership profiles with land use, urban functions and spatial characteristics to understand the different roles played by stations or areas within wider urban activity systems. Gan et al. (2020), for example, use metro-station ridership profiles to examine urban mobility patterns, while Zhang et al. (2021) use smart-card data to study changes in London's urban structure. Other research combines transport records with household travel surveys, socio-demographic information or points-of-interest data to investigate associations between mobility patterns and factors such as jobs-housing structure, population characteristics

and likely activity purposes (Long and Thill, 2015; Zhang et al., 2019; Sari Aslam et al., 2021; Yang et al., 2023).

This approach is particularly relevant to night-time public transport. Differences in night-time activity may appear in the timing of evening ridership, the persistence of activity after midnight, changes in directional flow and differences between weekdays and weekends. These patterns occur within areas that also differ in employment, residential, land-use and socio-demographic characteristics. McArthur et al. (2019) and Smeds et al. (2020) examine London's night-time transport from temporal and socio-spatial perspectives, while the London Night Workers Classification developed by Mavrogeni et al. (2025) provides a framework for comparing the geography of night-time workers across London. Together, these studies provide a basis for interpreting temporal public transport patterns in relation to the spatial structure of London's night-time activities and workforce.

2.4 Synthesis: Research Gap and Thesis Positioning

From current review, three linked strands of research frame this dissertation. Research on the urban night shows that night-time mobility is produced by multiple forms of activity and by transport conditions that vary across time and space (Schwanen et al., 2012; Van Liempt et al., 2015; Shaw, 2022). Smart-card research demonstrates that temporal transport records can be organised into recurring usage profiles (Briand et al., 2016; El Mahrsi et al., 2017; Gan et al., 2020). A related body of work shows that such profiles can be interpreted alongside land-use, employment and socio-spatial information. Taken together, these strands establish a pathway from night-time transport records to temporal typology and external area contextualisation, but they also reveal three connected gaps.

First, although research on the urban night has examined labour, accessibility, service provision and transport equity, it has less often identified recurring types of observed public-transport use within a dedicated night-time analytical window. Smart-card studies, by contrast, have mainly examined all-day travel, routine commuting, peak periods or general station functions, leaving

night-time activity embedded within the wider daily cycle.

Second, comparative attention to the temporal, directional and spatial organisation of night-time Rail and Bus use remains limited. Because the two modes differ in their data structures and operational characteristics, this gap calls for separate mode-specific analyses followed by cautious descriptive synthesis rather than a pooled or formally equivalent comparison.

Third, existing applications provide limited evidence on how transport-derived night-time typologies correspond to an independently constructed geography of night work and to wider socio-spatial conditions. This limits the interpretation of where different observed-use patterns occur and how they are embedded within the urban night.

The dissertation addresses the three gaps in sequence: it identifies night-time usage types separately for Rail stations and bus-served LSOAs; characterises their temporal, directional, day-type and spatial structures and their descriptive cross-mode differences; and examines their associations with the London Night Workers Classification and broader area context. These contextual associations are used to interpret the environments in which the patterns occur, not to infer individual passenger identities, travel purposes, causal mechanisms, unmet demand or service deficiency.

The next chapter describes the data and methods used to implement this research framework.

Chapter 3

Methodology

3.1 Research Design and Analytical Framework

The study adopts a sequential quantitative framework in which night-time public transport usage types are first identified from transport data and subsequently interpreted in relation to external area-level contexts.

First, temporal profiles were constructed separately for Rail stations and bus-served LSOAs. Rail profiles were based on station entries and exits, while Bus profiles were based on area-level boardings and estimated alightings. The two modes were analysed separately because their source data, temporal resolutions, spatial units and feature constructions differ.

Second, Gaussian Mixture Models were used to identify night-time usage patterns. The resulting cluster solutions were evaluated using fitted-model evidence, stability diagnostics, separation measures and the substantive interpretability of their temporal profiles.

After cluster assignment, activity volume, directional balance, late-night persistence and other behavioural descriptors were used to characterise the fixed clusters; these descriptors were not used to redefine cluster membership.

Third, the fixed cluster solutions were examined in relation to the London Night Workers Classification and 20 continuous urban background indicators. These contextual variables provided an external interpretive layer for assessing whether different observed-use types occurred within different night-work and socio-spatial environments.

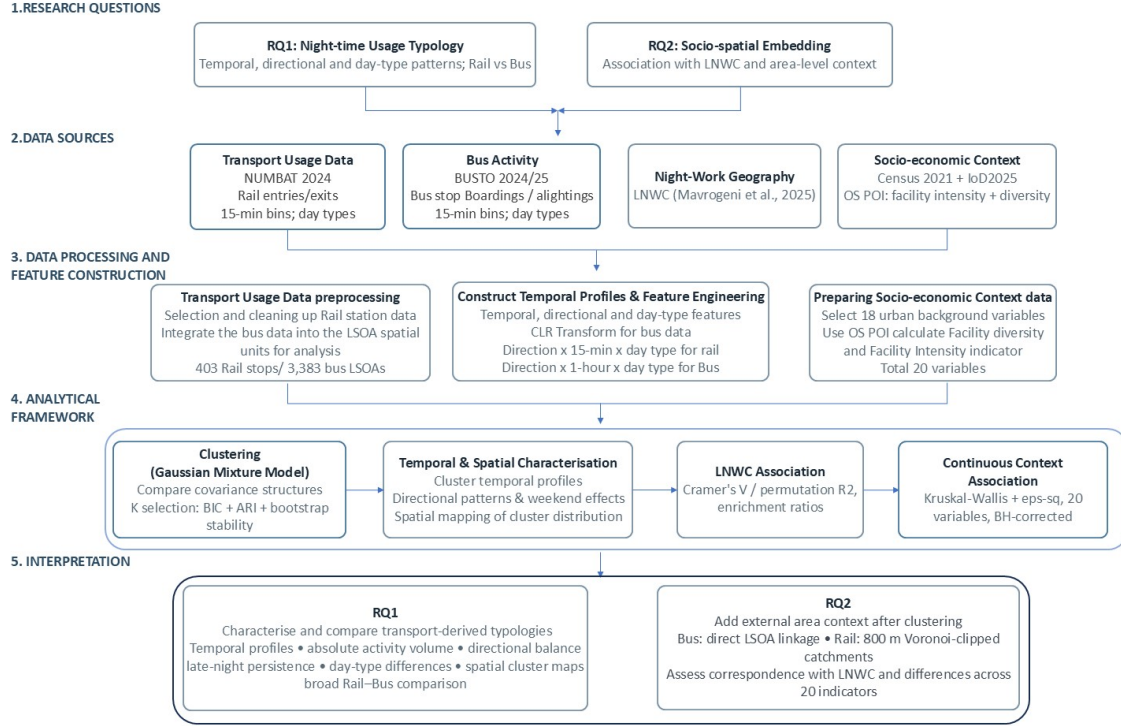


Figure 3.1: Research framework and structure

3.2 Data and Preparation

3.2.1 Rail Passenger Activity: NUMBAT 2024

Rail passenger activity was derived from Transport for London’s NUMBAT 2024 dataset (Transport for London, 2024). NUMBAT reports station entries and exits at 15-minute intervals for five representative day types within a typical operational week in autumn 2024.

Each operational day runs from 05:00 to 04:59 on the following calendar day. The dataset covers the London Underground, Docklands Light Railway, London Overground and Elizabeth line.

NUMBAT represents typical-day activity rather than continuous observations for every operating date. So, the data capture representative temporal patterns but do not describe day-to-day disruption or seasonal variation.

The raw data contained 471 station codes, some of which represented separate entry/exit systems for the same interchange rather than distinct locations. Twenty-nine such codes were merged into 14 co-located station units by aggregating activity across day type, direction and time interval, resulting in 456 physical stations. These were matched to NaPTAN (*National Public Transport Access Nodes*) records; 16 without a Greater London match and 37 tram-only units with no recorded activity were excluded. The final Rail dataset comprised 403 spatially referenced stations.

A fixed night-time window of 18:00–05:00 was applied to all five-day types. This produced 44 quarter-hour observations per day type and direction. Intervals following the end of service were retained as zero values rather than removed, maintaining a common feature length across day types while preserving the limited post-01:00 activity recorded on non-Night-Tube days by services such as the Elizabeth line and London Overground.

Each Rail station was therefore represented by 440 normalised features: 220 quarter-hour shares for entries and 220 for exits. Entry and exit profiles were retained separately for subsequent feature construction and clustering.

3.2.2 Bus Passenger Activity: BUSTO 2024/25

Bus passenger activity was obtained from TfL’s BUSTO 2024/25 Total Demand dataset (Transport for London, 2025). BUSTO reports passenger activity by route, direction, StopPoint and 15-minute interval for three representative day types: Weekday, Saturday and Sunday. Boarding values are derived from recorded passenger activity, whereas alighting values are estimated by TfL. The data should also be interpreted as representative measures of bus activity.

The release comprised 12 route-group and day-type files covering numbered and letter-prefix

bus routes. Unlike NUMBAT, BUSTO does not distinguish Friday from other weekdays, so Friday-night activity could not be analysed separately.

Because individual StopPoints often recorded low night-time flows and unstable temporal profiles, Bus activity was aggregated to 2021 LSOAs to provide a more stable spatial unit for temporal profiling and clustering. Before spatial assignment, StopPoints were first grouped using the official NaPTAN StopArea hierarchy, while those without a parent StopArea were kept as individual units. Passenger activity was then summed within each StopArea or individual StopPoint before assignment to LSOAs. This reduced 19,589 StopPoints to 12,007 spatial units. Of these, 10,879 were located within Greater London and contributed positive night-time activity to 3,797 LSOAs.

The records were then reconstructed into a continuous analytical night from 18:00 to 05:00. Because the source files are organised by calendar day, evening and post-midnight observations were recombined to represent each night: weekday evenings were paired with weekday post-midnight records, Saturday evenings with Sunday post-midnight records, and Sunday evenings with weekday post-midnight records. Following TfL's recommendation (Transport for London, 2025), the original 15-minute observations were subsequently aggregated to 11 hourly intervals per day type and direction, reducing the influence of very small quarter-hour values on temporal profiling.

A minimum-activity threshold was applied because very low-volume LSOAs produced sparse and unstable directional profiles. Following the low-activity screening principle used by Mariñas-Collado et al. (2022), an analogous threshold was adapted to the present data structure. Each LSOA was required to record at least 33 passenger activities ($11 \text{ hourly intervals} \times 3 \text{ day types}$) separately in both the boarding and alighting directions, equivalent to an average of at least one activity per hourly component in each direction. This retained 3,383 of the 3,797 LSOAs with positive night-time activity.

Each retained LSOA was represented by 66 temporal observations: $11 \text{ hourly intervals} \times 3 \text{ day types} \times 2 \text{ directions}$. The final Bus analytical dataset comprised 3,383 LSOAs with hourly

boarding and estimated-alighting profiles for subsequent feature construction and clustering.

3.2.3 Urban Night Contextual Data and Preparation

To interpret observed transport-use patterns in relation to their surrounding urban context, the study used the London Night Workers Classification (LNWC) alongside area-level socio-economic, demographic and functional indicators.

3.2.3.1 London Night Workers Classification

The London Night Workers Classification (LNWC) was used as an independent area-context layer for interpreting the transport clusters. Developed by Mavrogeni, Cheshire, Iliev and van Dijk as part of the Data After Dark programme (Mavrogeni et al., 2025), the LNWC combines mobile-phone footfall data with business-location data from the Directory of London Businesses. Businesses are classified using SIC codes to identify sectors associated with night-time work, and London LSOAs are grouped into seven area types representing different night-working contexts. Because the LNWC was constructed independently of the smart-card transport profiles analysed here, it provides an external reference for examining whether different transport-use types occur within different night-work contexts.

LNWC	Short name	Night-worker presence	Night-time-economy business presence
1	Central night-work hubs	Much higher than average, all time windows, peaking 18:00–21:00 weekdays	Much higher than average across all categories (arts/recreation, hospitality, retail dominant)
2	Inner suburbs, high night business	Higher than average	Above average across most categories
3	Inner suburbs, low night business	Higher than average (comparable to LNWC 2)	Below average; workers concentrated outside the tracked NTE categories (e.g. airport operations, building cleaning)
4	Suburban business, average activity	Average	Above average, led by security, hospitality, transport and storage
5	Suburban transport & health, low activity	Lower than average	Broadly average; transport, storage and health slightly above
6	Low-activity suburban	Lower than average (comparable to LNWC 5)	Below average across all categories
7	Night-worker periphery	Extremely low, especially 03:00–06:00 weekdays	Below average across all categories

Table 3.1: LNWC cluster types (adapted from Mavrogeni et al., 2025)

London Night Workers Classification (LNWC) by LSOA

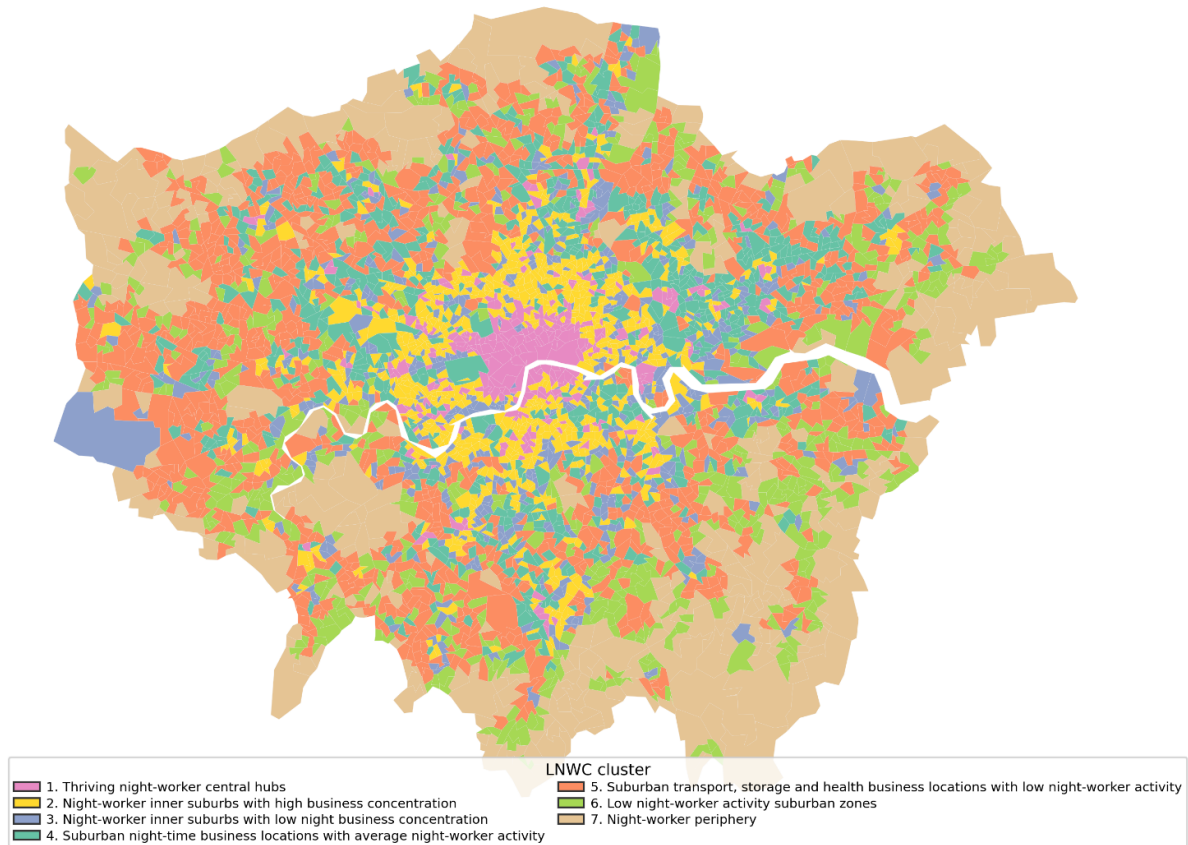


Figure 3.2: Spatial distribution of the seven London Night Workers Classification (LNWC) types across Greater London. Colours follow Mavrogeni et al. (2025).

3.2.4 Socio-economic, Demographic and Functional Indicators

The socio-economic and demographic contextual analysis draws on two main sources: the 2021 UK Census provided by the Office for National Statistics (ONS) (Office for National Statistics, 2021) and the English Indices of Deprivation 2025 (IoD2025) (Ministry of Housing, Communities and Local Government, 2025). The LSOA was retained as the common source geography to ensure spatial consistency across the contextual indicators. Based on the relevant literature, 18 variables were selected to represent different dimensions of local context, including deprivation, housing, car ownership, demographic structure, and residence-based employment in night-time-economy industries. The specific variables are summarised in the following table.

The selection of these variables was informed by previous studies examining socio-economic and demographic characteristics associated with human mobility patterns (Long and Thill, 2015; Cao and Hickman, 2019; Zhang et al., 2019; Sari Aslam et al., 2021; Yang et al., 2023). Residence-based employment in night-time-economy industries was defined with reference to the SIC-based industry categories used in the LNWC (Mavrogeni et al., 2025).

Table 3.2: Socio-economic and night-time employment variables

Domain	Variable	Source	Description
Area Deprivation	Education Skills and Training score	IoD2025	LSOA domain deprivation score for education and skills
Area Deprivation	Health Deprivation and Disability score	IoD2025	LSOA domain deprivation score for health and disability
Area Deprivation	Deprived in ≥ 1 dimension (%)	Census 2021, TS011	Share of households deprived in at least one of four Census deprivation dimensions
Housing & Car Ownership	Social rented (%)	Census 2021, TS054	Share of households in social rented tenure
Housing & Car Ownership	Private rented (%)	Census 2021, TS054	Share of households in private rented tenure
Housing & Car Ownership	No car or van (%)	Census 2021, TS045	Share of households with no car or van available
Household & Demographic Structure	Dependent children (%)	Census 2021, TS003	Share of households with dependent children
Household & Demographic Structure	Population density	Census 2021, TS001	Usual residents per km ²
Ethnicity	Asian ethnicity (%)	Census 2021, TS021	Share of usual residents identifying as Asian ethnic group
Ethnicity	Black ethnicity (%)	Census 2021, TS021	Share of usual residents identifying as Black ethnic group

(continued on next page)

(Table 3.2 continued)

Domain	Variable	Source	Description
Economic Activity	Unemployed (%)	Census TS066	Share of economically active residents aged 16+ who are unemployed
Night-time Industry (residence)	Accommodation & food services (%)	Census TS060A	Share of employed residents (16+) working in hospitality-related industries
Night-time Industry (residence)	Transport & storage (%)	Census TS060A	Share of employed residents working in transport and storage industries
Night-time Industry (residence)	Administrative & support services (%)	Census TS060A	Share of employed residents working in administrative/support-service industries (proxy for security-related occupations)
Night-time Industry (residence)	Wholesale & retail trade (%)	Census TS060A	Share of employed residents working in wholesale and retail industries
Night-time Industry (residence)	Human health & social work (%)	Census TS060A	Share of employed residents working in health and social-care industries
Night-time Industry (residence)	Manufacturing (%)	Census TS060A	Share of employed residents working in manufacturing industries
Age structure	Aged 20–34 (%)	Census TS007b	Share of residents aged 20 to 34
Environment	POI count	Ordnance Survey Points of Interest	Total number of points of interest within the LSOA

(continued on next page)

(Table 3.2 continued)

Domain	Variable		Source		Description		
Environment	POI	Diversity	Ordnance	Sur-	Shannon	diversity	index
	(Shannon H)		vey Points of In-		across the nine top-level OS		
			terest		POI groups		

Ordnance Survey Points of Interest (POI; June 2026 release, accessed through EDINA Digimap) (Ordnance Survey, 2026) were used to characterise the functional environment of each area. Drawing on previous studies, two indicators were derived from the POI data: facility intensity and functional diversity (Peiret-García et al., 2026).

Facility intensity was measured as the total number of POIs within each LSOA and was transformed using $\log(1 + x)$ to reduce the influence of the strongly right-skewed count distribution. Functional diversity was measured using the Shannon index across the nine top-level OS POI groups:

$$H = - \sum_{g=1}^G p_g \ln(p_g), \quad (3.1)$$

where p_g is the proportion of POIs belonging to group g . Larger values indicate a more even distribution across facility groups.

Together, the 18 Census and deprivation indicators and the two POI measures produced a set of 20 continuous contextual variables. These were calculated at LSOA level and linked directly to Bus units or aggregated across Rail station catchments using the procedure described in Section 3.3.4.1.

3.3 Analytical Methods

3.3.1 Feature Engineering

Following data preparation, the Rail and Bus temporal profiles were converted into feature spaces suitable for clustering. Feature construction was designed to emphasise differences in the temporal shape of public transport activity rather than differences in absolute passenger volume.

3.3.1.1 Direction-specific Normalisation

Night-time passenger activity is highly uneven across London, with major interchanges and central locations recording substantially greater volumes than most other units. Clustering raw passenger counts might risk grouping locations primarily according to activity scale rather than according to when they are used.

To focus the classification on temporal structure, activity was normalised separately for each unit and direction across the complete analytical period, following approaches used in previous temporal mobility research (Briand et al., 2016; El Mahrsi et al., 2017; Verma et al., 2021).

For unit i , direction d and time component t , the normalised share was:

$$p_{i,d,t} = \frac{c_{i,d,t}}{\sum_{t'} c_{i,d,t'}}, \quad \sum_t p_{i,d,t} = 1 \quad \forall i, d \quad (3.2)$$

where $c_{i,d,t}$ is the recorded or estimated activity and the denominator is the unit's total activity in direction d across the complete analytical period.

Rail entries and exits were normalised separately, as were Bus boardings and alightings.

Each directional profile entered the clustering feature space independently, limiting the influence of differences in absolute directional volume on cluster assignment.

Absolute passenger activity was retained separately and reintroduced after clustering as a behavioural

descriptor.

3.3.1.2 Compositional Transformation of Bus Features

Direction-specific normalisation produces compositional profiles in both modes because the shares within each direction sum to one and cannot vary independently.

An increase in one interval necessarily reduces the shares available to the others, so ordinary Euclidean distances may distort their relative structure (Aitchison, 1982). The adopted Bus model addresses this constraint through a CLR transformation of its 33 hourly components. Rail retains the normalised raw-share representation across 220 quarter-hour components per direction as a pragmatic modelling choice; the remaining compositional constraint is recognised as a limitation rather than assumed to be negligible.

To represent relative differences between intervals in an unconstrained coordinate space, the adopted Bus specification applied a centred log-ratio (CLR) transformation separately to the boarding and alighting profiles.

Because the CLR transformation requires strictly positive components, zero values were first replaced using a prior-weighted Bayesian-multiplicative procedure (Martín-Fernández et al., 2015).

For LSOA i , direction d and concatenated day-type–hour component t , the network-wide temporal prior was defined as:

$$\pi_{dt} = \frac{\sum_i c_{idt}}{\sum_i \sum_{t'} c_{idt'}}. \quad (3.3)$$

The zero-adjusted composition was then calculated as:

$$\tilde{p}_{idt} = \frac{c_{idt} + \alpha \pi_{dt}}{\sum_{t'} c_{idt'} + \alpha}, \quad \alpha = 1. \quad (3.4)$$

Thus, one notional observation was distributed across the 33 temporal components in proportion to

the corresponding network-wide activity shares. This preserved the observed temporal structure of the network rather than replacing every zero with the same arbitrary constant.

The adjusted composition was transformed as:

$$z_{idt} = \ln\left(\frac{\tilde{p}_{idt}}{g(\tilde{\mathbf{p}}_{id})}\right), \quad g(\tilde{\mathbf{p}}_{id}) = \left(\prod_{t=1}^D \tilde{p}_{idt}\right)^{1/D}, \quad (3.5)$$

or equivalently,

$$z_{idt} = \ln(\tilde{p}_{idt}) - \frac{1}{D} \sum_{s=1}^D \ln(\tilde{p}_{ids}). \quad (3.6)$$

The transformed components sum to zero within each direction, and Euclidean distance in CLR coordinates corresponds to Aitchison distance between the original compositions.

Mode	Analytical unit	Direction	Day types	Time resolution	Components per direction	Transformation	Final feature matrix
Rail	403 stations	Entry / Exit	MON, TWT, FRI, SAT, SUN	15 min, 18:00–05:00	220	Direction-specific shares	normalised 403×440
Bus	3,383 LSOAs	Boarding / Alighting	Weekday, SAT, SUN	1 h, 18:00–05:00	33	Normalisation $\rightarrow$ zero replacement $\rightarrow$ CLR	$3,383 \times 66$

Table 3.3: Structure of the Rail and Bus feature matrices used for GMM clustering

3.3.2 Gaussian Mixture Model Clustering

3.3.2.1 Model Rationale

Gaussian Mixture Models (GMMs) were used to identify locations with similar temporal usage profiles, following their application to transport data at station level (Kim et al., 2017; Verma et al., 2021; Wang et al., 2026) and passenger level (Briand et al., 2016; El Mahrsi et al., 2017). A GMM models the observed profiles as a mixture of K multivariate Gaussian components, each defined by its own mean vector and covariance structure. For a profile x ,

$$p(x) = \sum_{k=1}^K \phi_k \mathcal{N}(x | \mu_k, \Sigma_k), \quad \sum_{k=1}^K \phi_k = 1, \quad (3.7)$$

where ϕ_k is the mixing weight of component k , while μ_k and Σ_k are its mean vector and covariance matrix, respectively. The probability that unit i belongs to component k is

$$\gamma_{ik} = \frac{\phi_k \mathcal{N}(x_i | \mu_k, \Sigma_k)}{\sum_{j=1}^K \phi_j \mathcal{N}(x_i | \mu_j, \Sigma_j)}. \quad (3.8)$$

Each unit was assigned to the component with the highest posterior probability, and these probabilities also indicate how clearly each observation is associated with its component.

This specification is well suited to transportation data because temporal profiles may differ in both shape and variation across locations. Unlike k-means, which favours approximately spherical clusters, a GMM allows each component to have its own covariance structure. The fitted components are interpreted as analytical typologies of temporal activity rather than naturally occurring groups.

3.3.2.2 Covariance Structure and Model Fitting

Models were estimated by Expectation-Maximisation using scikit-learn. A small regularisation term was added to the covariance diagonal so that the high-dimensional feature spaces could be fitted stably.

Expectation-Maximisation can settle on different local optima from different starting values. So, each candidate specification was fitted from multiple initialisations, retaining the solution with the highest likelihood.

The covariance structure was treated as an empirical modelling choice rather than fixed in advance. Spherical, tied, diagonal and full specifications were compared using the Bayesian Information Criterion (BIC) (Schwarz, 1978; Fraley and Raftery, 2002).

All four covariance structures were carried forward into the model comparison so that covariance form and K were evaluated together within each mode. The selected mode-specific specifications are reported with the Results in Section 4.1, where they form part of the evidence supporting the final cluster solutions.

3.3.2.3 Selection of the Number of Clusters

Selecting K required an analytical judgement between competing representations of the same mode-specific data. A prespecified range of K values and covariance structures was therefore compared within each mode. The adopted solution had to combine competitive fitted-model evidence with reproducibility and a temporal profile structure that remained substantively interpretable.

BIC provided the principal starting point for that comparison. It weighs fitted likelihood against the number of free parameters, so an additional component is preferred only when its improvement in fit justifies its added complexity:

$$BIC = -2\ln(\hat{L}) + m\ln(n), \quad (3.9)$$

where $\hat{L}$ is the maximised likelihood, m is the number of estimated parameters and n is the number of units. Lower BIC values indicate a better balance between model fit and complexity.

BIC established the statistically competitive solutions, after which reproducibility was examined across random initialisations using the Adjusted Rand Index (ARI) (Hubert and Arabie, 1985) and under bootstrap resampling. ARI measures agreement between partitions after accounting for chance, while the bootstrap tests whether the same structure reappears when the sample is perturbed. These checks qualify the

fitted-model evidence by showing how consistently the proposed typology can be recovered.

Silhouette and Davies-Bouldin indices supplied a further reading of separation and compactness (Davies and Bouldin, 1979; Rousseeuw, 1987).

Read alongside the temporal profiles and descriptor summaries, they showed whether the BIC-preferred solution yielded a typology with a coherent empirical interpretation.

3.3.3 Post-clustering Behavioural Descriptors

Normalisation allowed the clustering to focus on temporal shape, but it intentionally removed absolute activity scale and the relative magnitude of the two directions.

These features remain analytically important, so they were recovered after assignment as descriptors calculated from the unnormalized activity data and were not used as clustering inputs.

For visual comparison, cluster means were also standardised against the relevant mode-wide mean; the resulting z-score panels are descriptive summaries of the fixed-cluster profiles.

The following table presents the continuous descriptive variables that we used.

Table 3.4: Post-clustering behavioural descriptor variables

Dimension	Metric and mode	Operational definition	Interpretation
Activity scale	<code>log_total_activity</code> — Rail and Bus	$\log(1 + A_i)$ where A_i is total activity summed across all included day types, directions and intervals	Restores the absolute activity scale removed by normalisation. Larger values indicate higher total observed or estimated activity.

(continued on next page)

(Table continued)

Dimension	Metric and mode	Operational definition	Interpretation
Directional balance	direction_balance — Rail and Bus	$B_i = (O_i - A_i)/(O_i + A_i)$, where O_i denotes entries for Rail or boardings for Bus, and A_i denotes exits or alightings	Values near zero indicate directional balance. Positive values indicate a stronger departure/origin-oriented pattern; negative values indicate a stronger arrival/destination-oriented pattern.
Later-night continuation	post_2300_share — Rail	Activity during 23:00–05:00 divided by total activity during 18:00–05:00	Measures the proportion of Rail activity occurring after 23:00.
Later-night continuation	post_2300_share — Bus	Activity during 23:00–05:00 divided by total activity during 18:00–05:00	Measures the proportion of Bus activity occurring after 23:00.
Persistence relative to the evening	post_midnight_persistence — Bus only	Post-midnight activity (00:00–05:00) divided by early-evening activity (18:00–21:00)	Distinguishes LSOAs whose activity remains comparatively strong after midnight from those that are simply busy throughout the evening.

(continued on next page)

(Table continued)

Dimension	Metric and mode	Operational definition	Interpretation
Week structure	weekend_to_ weekday_ratio — Rail	Mean activity across Saturday and Sunday divided by mean activity across Monday and Tuesday–Thursday, computed over an 18:00–01:00 window. Friday is excluded from both groups	Values above one indicate relatively stronger weekend use; values below one indicate a profile more strongly associated with weekdays. The narrower window reflects an earlier data constraint and is retained for continuity; a sensitivity check confirmed the resulting cluster-level pattern is not materially affected by widening it to the full night window.
Week structure	weekend_to_ weekday_ratio — Bus	Mean activity across Saturday and Sunday divided by mean weekday activity, computed over the full 18:00–05:00 window. Friday cannot be separated from weekday at source and is therefore included within the weekday reference	Values above one indicate relatively stronger weekend use; values below one indicate a profile more strongly associated with weekdays.

3.3.4 Post-clustering Association with LNWC and Urban Background Indicators

Having completed the clustering of night-time public transport usage patterns, the analysis follows the research framework to relate the fixed clusters to LNWC and the wider city background variables.

LNWC provides a categorical account of night-work context; the urban background indicators describe socio-economic, demographic and functional conditions. Both comparisons are conducted after clustering, at the station-catchment or LSOA scale, to test for area-level correspondence with the observed-use typology.

3.3.4.1 Mode-specific Spatial Linkage

To relate the fixed clusters to area context, Rail and Bus were linked separately to reflect their different spatial units.

For Bus

Because Bus activity was already aggregated to 2021 LSOAs before clustering, each fitted LSOA could be matched directly to its LNWC category and continuous contextual indicators using the LSOA code.

For Rail

Rail required an explicit areal linkage because stations are represented as points, whereas the contextual variables describe areas. Although Euclidean buffers provide a clear distance-based definition of station catchments, they generate substantial overlap in station-dense central London, causing the same area to be assigned to multiple stations.

To address this, the study adopted a hybrid buffer–Voronoi approach previously applied in related research (Lai et al., 2017). An 800-metre buffer, representing approximately a 15-minute walk (Peiret-García et al., 2026), was generated around each of the 403 stations and clipped by a Voronoi tessellation of the same station locations. The Voronoi boundaries assign overlapping space to the nearest station, while the buffer retains a common maximum catchment distance, producing mutually exclusive station catchments.



Figure 3.3: The London rail stations with their respective hybrid Voronoi-clipped catchment zones

Each catchment was intersected with the 2021 LSOA boundaries, and duplicate geometry fragments were collapsed to make sure each distinct station–LSOA pair was represented once.

Of the 403 Rail stations included in the clustering, 389 had at least one eligible 2021 London LSOA intersecting their 800-metre Voronoi-clipped catchment and were retained for the contextual analyses. The remaining 14 stations were outer-network locations whose catchments did not intersect the spatial extent of the available London LSOA contextual data; they were excluded from the contextual analyses only and remained part of the clustering.

Station context was defined as the distribution of LSOA-level conditions among the distinct LSOAs intersecting the 800-metre Voronoi-clipped catchment. Each intersecting LSOA received equal weight, irrespective of its population or the proportion of its area falling within the catchment.

For continuous social indicators, station-level values were calculated as the arithmetic mean of the intersecting LSOAs. LNW was summarised as a seven-part composition, representing the proportion of

intersecting LSOAs in each class. The same catchments and equal LSOA weighting were used in both cases.

3.3.4.2 Correspondence with the Geography of Night Work

The first contextual analysis assessed whether the fixed transport-use clusters align with the independently derived LNWC geography. Because Bus and Rail are defined at different spatial scales, mode-specific methods were used. Bus LSOAs each carry a single LNWC category, whereas Rail station catchments are represented as seven-part LNWC compositions because individual rail catchments often intersect multiple LSOAs with different LNWC types.

For Bus, both cluster membership and LNWC type are categorical variables at LSOA level. Their association was tested using Pearson’s chi-square test of independence, with effect size summarised by Cramér’s V (Cramér, 1999):

$$V = \sqrt{\frac{\chi^2}{n \cdot \min(r-1, c-1)}} \quad (3.10)$$

where χ^2 is the chi-square statistic, n is the number of Bus LSOAs, and r and c are the numbers of rows and columns in the contingency table.

For Rail, each of the 389 stations is described by a seven-part LNWC composition based on equal weighting of intersecting classified LSOAs. Differences in these compositions across fixed Rail clusters were evaluated using a label-permutation test. Cluster labels were randomly reassigned 999 times while station-level compositions were held constant. The observed R^2 , defined as the ratio of between-cluster to total compositional variation, was compared against this null distribution to test whether observed differences exceed those expected under random assignment (Anderson, 2001):

$$R^2 = \frac{SS_{between}}{SS_{total}} \quad (3.11)$$

To interpret the direction of association, enrichment ratios were calculated for each cluster and LNWC

type as the share of an LNWC type within a cluster divided by its share across the relevant eligible benchmark — the 3,383 eligible fitted Bus LSOAs, or the equal-station-weighted composition across the 389 eligible Rail stations:

$$ER_{kg} = \frac{s_{kg}}{s_{\cdot g}}. \quad (3.12)$$

A value above one indicates over-representation relative to that benchmark. Where a cluster showed one clearly strongest enrichment, that LNWC type was used as a concise contextual label.

3.3.4.3 Differences in Urban Background Indicators Across Clusters

The 20 continuous urban background indicators were analysed after clustering, using fixed cluster labels as grouping variables. These indicators were not used in cluster construction or refinement. The analysis instead examined whether contextual distributions differed across transport-use types. Rail and Bus were analysed separately, using 389 Rail station catchments and 3,383 Bus LSOAs.

For each indicator, a Kruskal–Wallis test assessed whether distributions differed across clusters within each mode (Kruskal and Wallis, 1952). This non-parametric test was used because the variables included bounded proportions and skewed counts that did not meet ANOVA assumptions. Benjamini–Hochberg correction was applied across the 20 tests separately for Rail and Bus (Benjamini and Hochberg, 1995).

ϵ^2 was used to rank the magnitude of between-cluster differentiation across the 20 indicators within each mode:

$$\epsilon^2 = \frac{H - k + 1}{n - k}, \quad (3.13)$$

where H is the Kruskal–Wallis statistic, k is the number of clusters and n is the number of units in the test.

To identify which clusters drove these differences, each cluster was compared with all remaining units using two-sided Mann–Whitney U tests. Rank-biserial correlation was used to describe direction and

magnitude. Benjamini–Hochberg correction was applied across 80 Bus comparisons (4 clusters $\times$ 20 indicators) and 100 Rail comparisons (5 clusters $\times$ 20 indicators).

Standardised cluster-mean z-scores were reported as descriptive summaries, indicating whether each cluster was above or below the mode-wide mean. Inference was based on the Kruskal–Wallis and Mann–Whitney results, their effect sizes, and adjusted p-values.

3.4 Ethical Considerations and Methodological Boundaries

This study used publicly available, aggregate transport and area-level contextual data. No personal identifiers, individual smart-card records or individual movement trajectories were accessed, and the research involved no participant recruitment or direct interaction with human subjects.

Socio-economic and demographic indicators were interpreted only as characteristics of the surrounding areas. They were not treated as attributes of individual passengers, and the analysis does not infer individual travel behaviour from area-level associations.

Generative AI tools were used in an assistive capacity for code development, debugging and language proofreading. All analytical decisions, interpretation and final written content remain the responsibility of the author.

Chapter 4

Results

This chapter presents the empirical results of the analysis of night-time public transport use in London. Following the analytical framework set out in Chapter 3, temporal profiles were clustered separately for 403 Rail stations and 3,383 bus-served LSOAs. The resulting types were first characterised in terms of their temporal, directional and spatial features. They were then examined in relation to the London Night Workers Classification (LNWC) and wider area context, using the 389 Rail stations with eligible contextual catchment coverage and all 3,383 Bus LSOAs.

4.1 Selection of the Cluster Solutions

Models with $K=2-12$ and alternative covariance structures were compared, with BIC used as the primary model-selection criterion. Separation and stability diagnostics were used to qualify, rather than determine, the selected solutions. The minimum BIC was obtained by a diagonal-covariance $K=5$ model for Rail and a full-covariance $K=4$ model for Bus, which were retained for subsequent analysis.

Stability diagnostics indicate that the retained solutions should be interpreted as analytical typologies rather than invariant groupings. Rail $K=5$ was highly reproducible across random initialisations (mean pairwise ARI = 0.964), but showed more moderate recovery under bootstrap resampling (mean ARI = 0.510), with the least reproducible cluster reaching a mean minimum matched Jaccard of 0.301. Bus $K=4$ showed stronger overall bootstrap agreement (mean ARI = 0.785), although its least stable component was similarly sensitive to resampling (mean minimum matched-cluster Jaccard = 0.316).

Rail model-selection diagnostics (n=403); red marker indicates adopted K=5. Panels use independent y-axes.

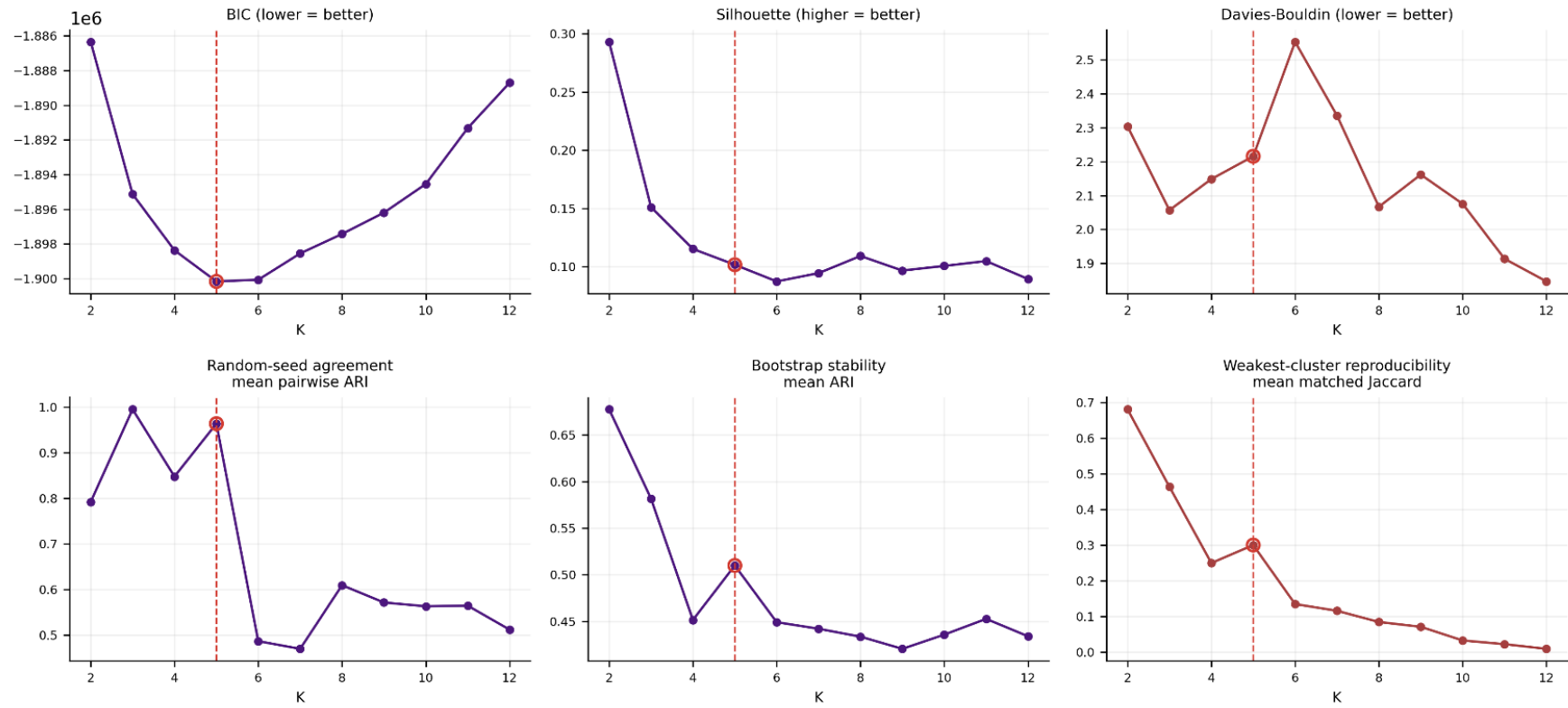


Figure 4.1: Rail K selection evidence

Bus model-selection diagnostics (n=3,383); red marker indicates adopted K=4. Panels use independent y-axes.

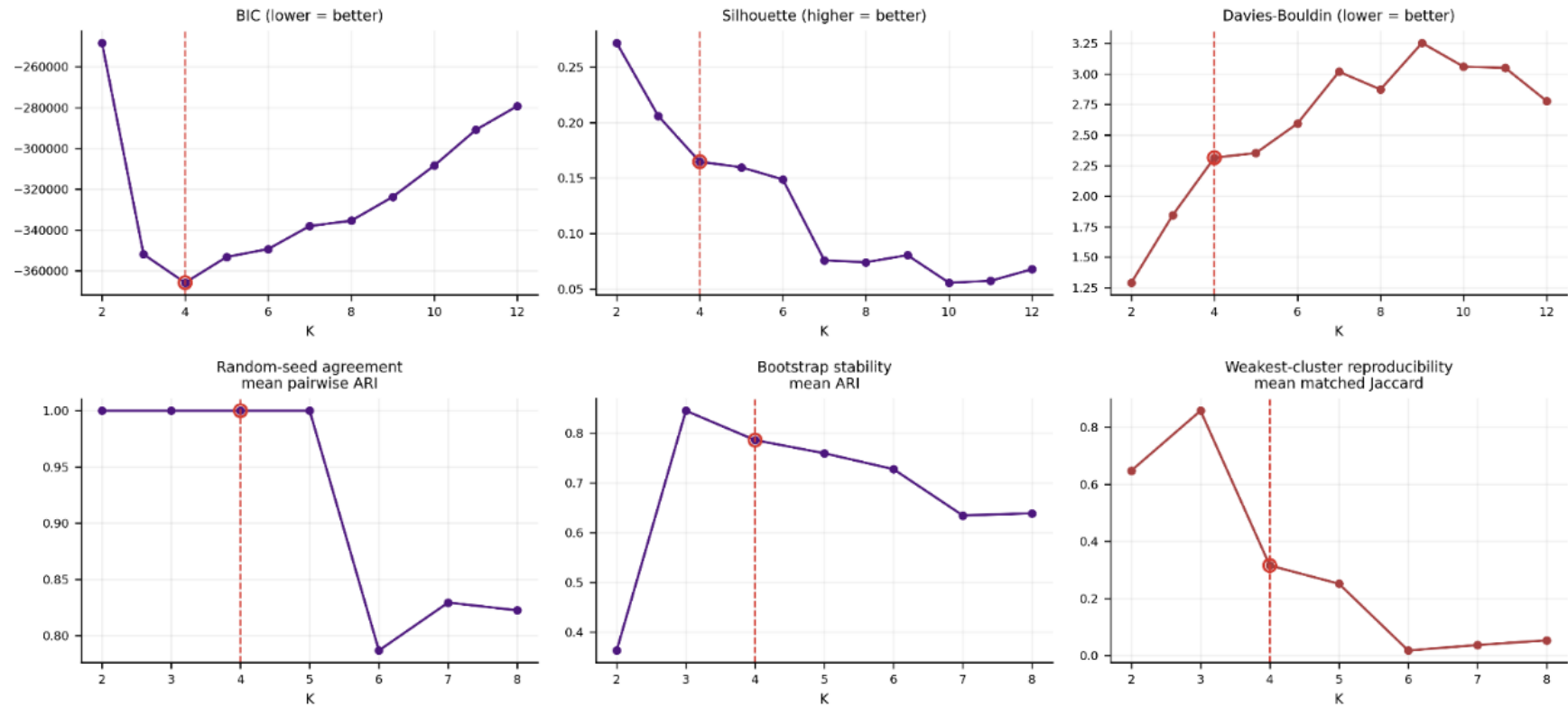


Figure 4.2: Bus K selection evidence

Conditional on the fitted models, assignment ambiguity was low: 98.5% of Rail stations and 99.0% of Bus LSOAs had maximum posterior probabilities of at least 0.99. These probabilities describe assignment certainty within the fitted solution and should not be interpreted as evidence of resampling stability.

4.2 Night-time Public Transport Usage Types

Following the selection of the mode-specific cluster solutions, this chapter presents the identified night-time public transport usage types through their observed temporal profiles, post-clustering descriptors and spatial distributions. Rail and Bus were examined separately throughout because their analytical units and feature constructions differ.

4.2.1 Five Night-time Usage Types: Rail System

The following figures present the mean temporal profiles and spatial distribution of the five Rail clusters, respectively. The sections below summarise the main observed characteristics of each cluster.

Rail station profiles by cluster: 18:00–05:00, n=403

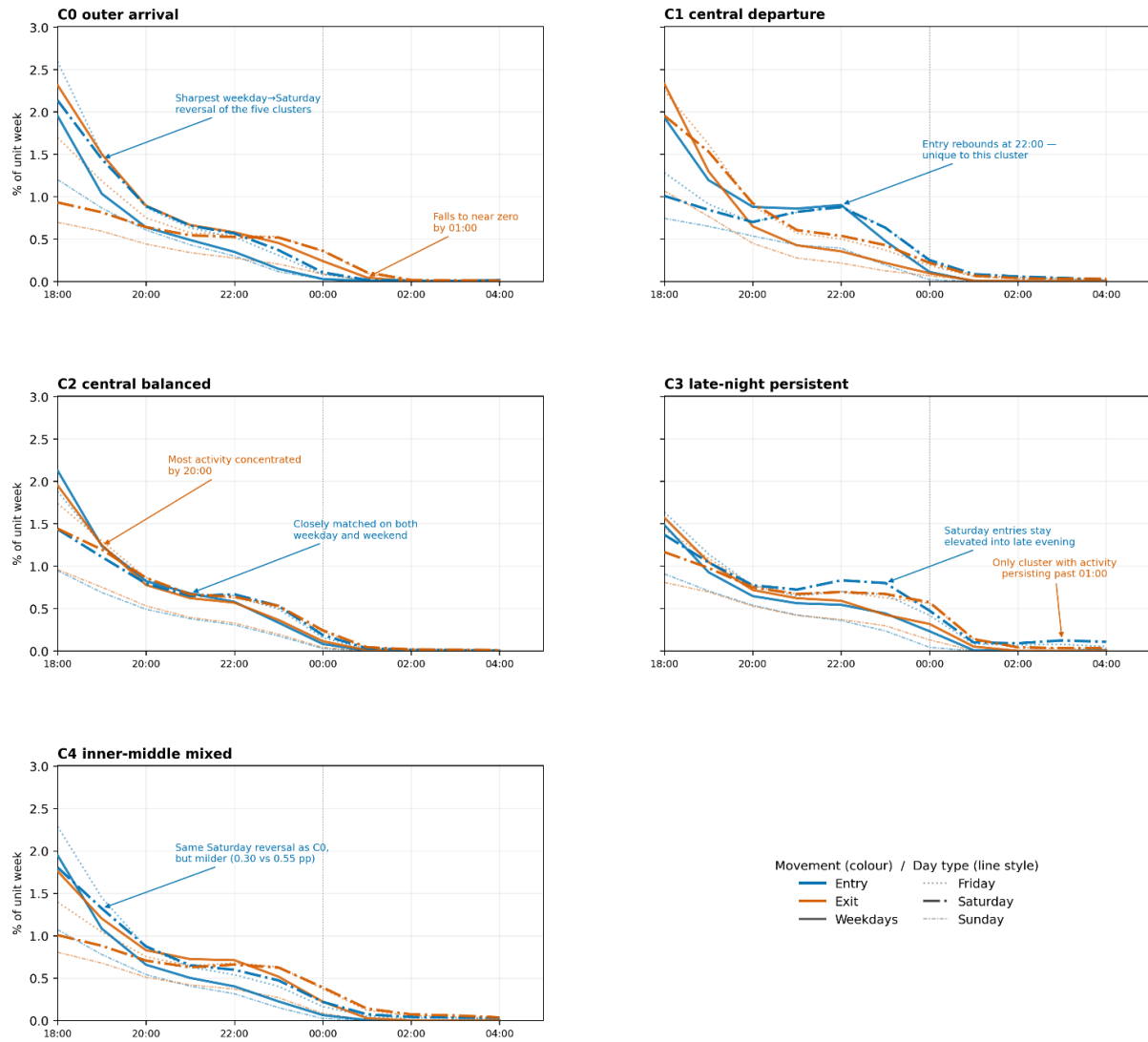


Figure 4.3: Rail station temporal profiles (403 stations; K=5; 18:00–05:00). Colour distinguishes entries and exits; line style distinguishes day types; panels, labels and titles identify clusters.

Rail Station Cluster Membership: $K = 5$
(403 stations, all-modes clustering population)

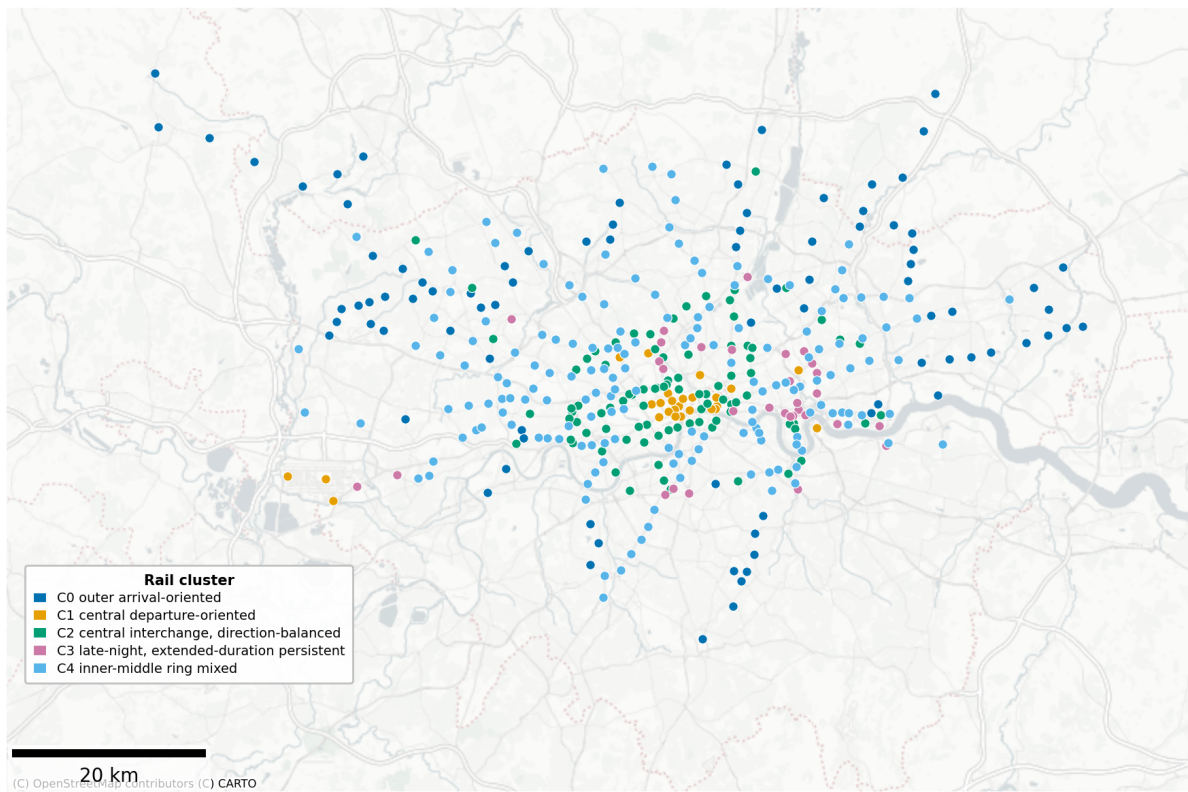


Figure 4.4: Rail cluster map: station-point analytical units

Post-clustering behavioural descriptors for the five Rail clusters.

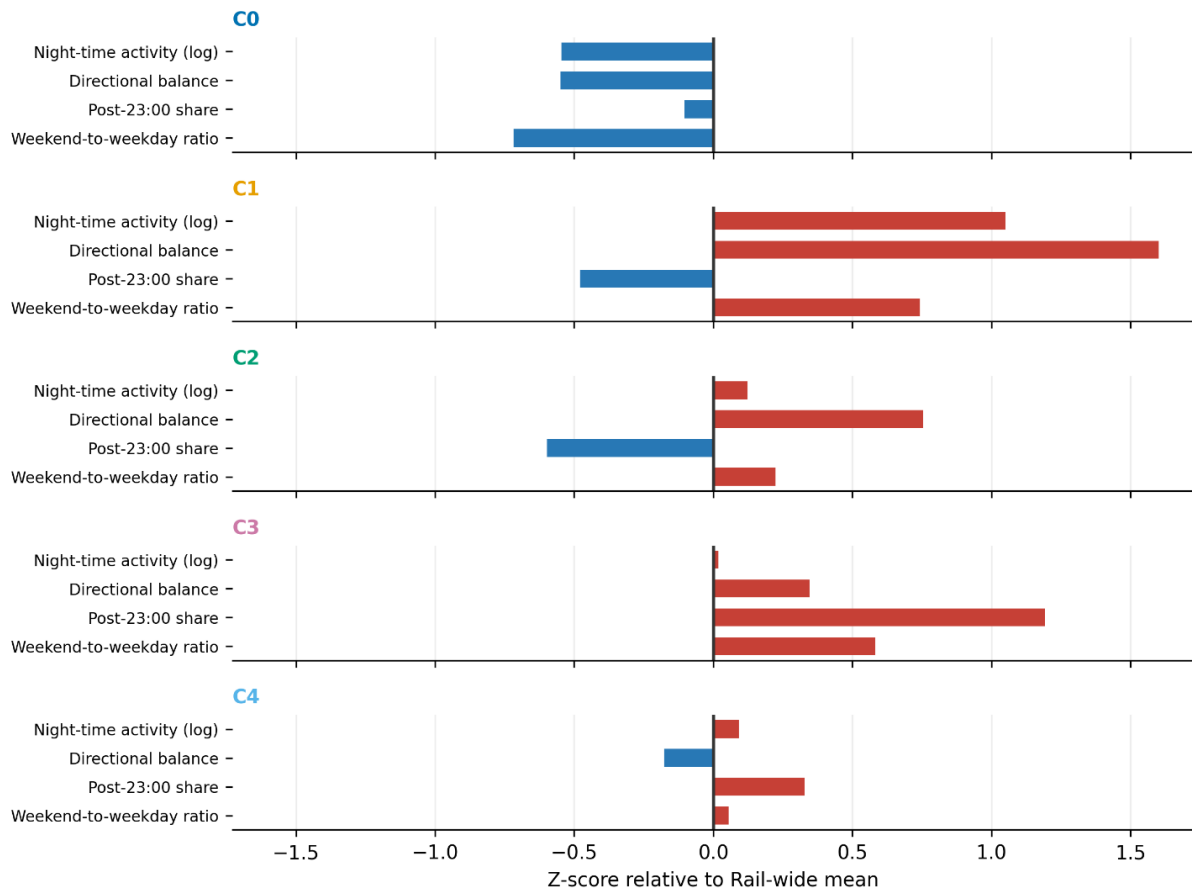


Figure 4.5: Post-clustering behavioural descriptors for the five Rail usage types. Values are cluster-mean z-scores relative to the Rail-wide mean and standard deviation. Positive and negative values indicate above- and below-average values, respectively; for directional balance, higher values indicate a stronger departure/entry orientation.

C1: Night-time activity centres cluster (central departure) (n = 26). C1 is tightly concentrated in central London, particularly around the West End and central activity areas. It includes major West End and central stations such as Oxford Circus and Piccadilly Circus, alongside a small number of more specialised transport nodes.

It records the highest overall night-time activity of all five clusters. The temporal profile shows strong exit activity between 18:00 and 20:00, followed by a distinctive rise in entries around 22:00. This later entry rebound is not visible to the same extent in the other clusters. The post-clustering descriptors also show the strongest positive directional balance, while the share of activity after 23:00 remains below the Rail

average.

C1 is thus characterised by high-volume central activity and a pronounced departure orientation later in the evening, rather than by sustained deep-night use.

C0: Outer arrival-oriented (n = 89). C0 forms the clearest outer-London Rail type. Its stations are widely distributed along radial corridors at the edge of the network, with a mean distance of 17.76 km from central London.

Activity is concentrated in the early evening and declines rapidly thereafter, reaching very low levels by around 01:00. On weekdays, exits exceed entries, giving the cluster a clear arrival orientation. The balance changes most strongly on Saturday, when entries become relatively more prominent earlier in the evening and exits later at night.

Among the five clusters, C0 also has the lowest overall activity and the weakest relative weekend activity.

The combination of peripheral location, low activity and weekday exit dominance makes C0 the clearest outer arrival-oriented Rail type.

C4: Inner–middle ring cluster (n = 167). C4 is the largest Rail cluster and forms a broad band across inner and middle London. Its stations extend along several rail corridors and occupy an intermediate spatial position between the central clusters and the more peripheral C0.

The temporal profile shares some features with C0, including a mild shift in directional balance on Saturday, but the change is less pronounced. Activity also persists later into the night than in C0, reflected in an above-average post-23:00 share. Overall volume remains close to the Rail mean, while directional balance shows only a modest arrival tendency.

C4 is therefore better described as an inner–middle mixed type than as a strongly arrival-oriented cluster.

C3: Late-night, extended-duration persistent (n = 31). C3 is a small cluster located mainly across inner and eastern London. Unlike the larger Rail groups, its defining feature is not overall activity volume but the persistence of activity later into the night.

Its temporal profiles decline more slowly after the early evening, with Saturday entries remaining elevated into the late evening. C3 also shows the clearest continuation beyond 01:00 and has by far the highest post-23:00 activity share in the post-clustering descriptors.

This distinguishes C3 from both high-volume central clusters and larger arrival-oriented groups, which may reflect a combination of nearby night-time activities, interchange functions, local demand, and late-running services.

C2: Central, direction-balanced cluster (n = 90). C2 covers central London and a number of major inner-London interchange stations, including King's Cross St Pancras, Liverpool Street and London Bridge.

The entry and exit profiles remain closely aligned across most day types. Most activity is concentrated before 20:00 and declines steadily through the rest of the evening.

The close alignment of entry and exit activity distinguishes C2 from the stronger directional patterns of C1 and C0, making it a relatively high-activity central type without a pronounced arrival or departure bias.

Overall Rail structure. The Rail analysis identifies five relatively distinct night-time station types with different spatial location and temporal usage characteristics.

Across the five clusters, Rail usage differs in spatial position, directional balance, activity scale, the timing of evening activity and the extent to which activity continues later into the night. The clearest spatial contrast is between the high-activity central departure pattern of C1 and the low-activity outer arrival pattern of C0, while other clusters occupy less strongly directional or more temporally persistent positions.

C1 is the busiest and most centrally located type, yet its share of activity after 23:00 falls below the Rail average. C3, by contrast, records lower overall activity but the strongest late-night continuation and is concentrated mainly across inner and eastern London. The Rail typology therefore captures several forms of night-time station use that would not be distinguished by activity volume or spatial position alone.

4.2.2 Four Night-time Area Profiles: Bus System

Following figures show the mean temporal profiles and spatial distribution of the four Bus clusters. Although clustering was performed on CLR-transformed features, boarding and estimated-alighting shares are shown on original scale for interpretation.

Compared with Rail, the Bus profiles are less sharply differentiated. All four clusters show relatively high activity around 18:00, followed by a gradual evening decline, low activity after midnight, and some recovery in boarding around 04:00.

The main differences are in how quickly activity declines, how long it persists into the night, directional balance and overall activity level. More persistent patterns are concentrated mainly in central and inner London.

Bus LSOA profiles by cluster: 18:00–05:00, n=3,383

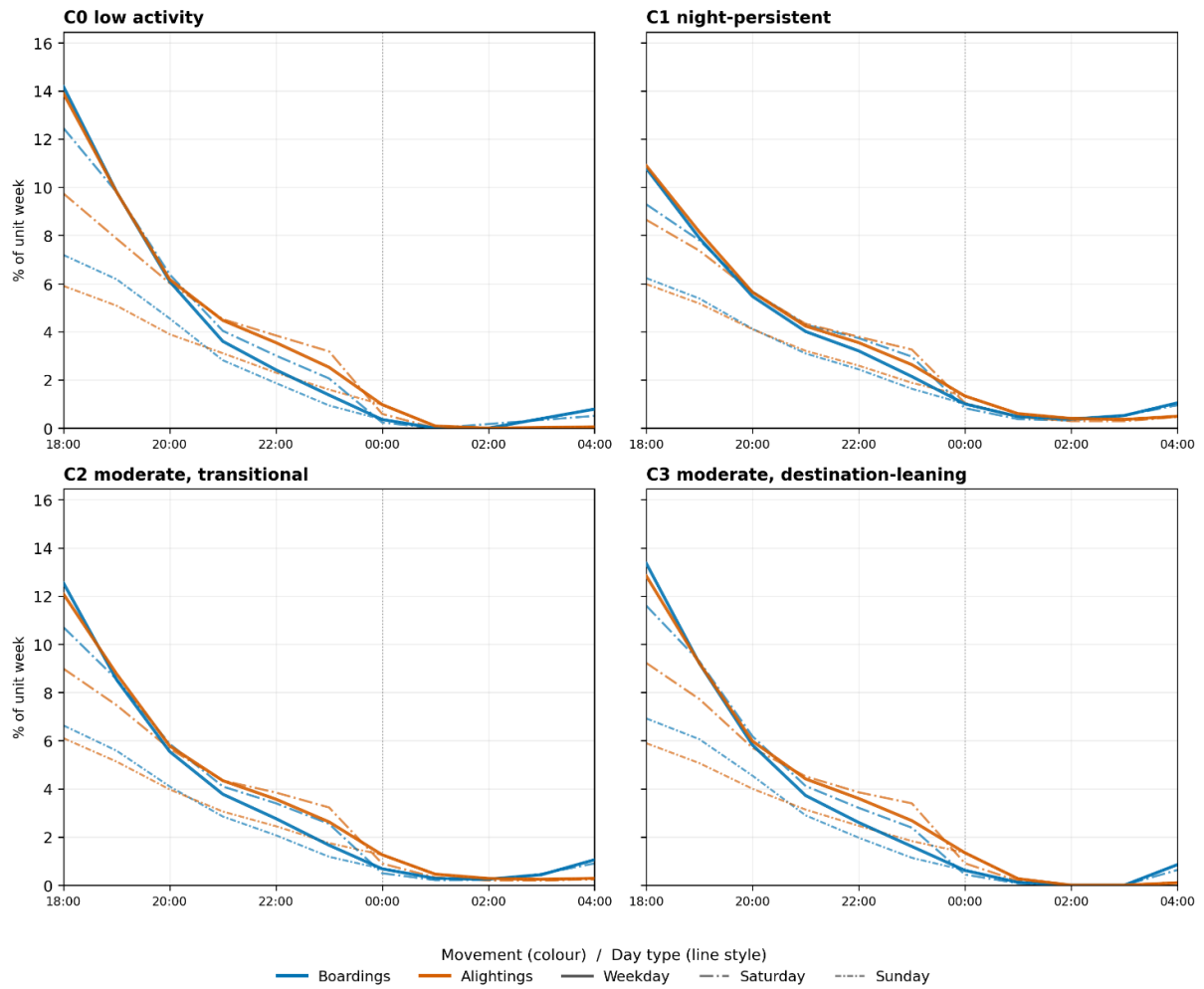


Figure 4.6: Bus temporal profiles (3,383 LSOAs; K=4; 18:00–05:00; threshold 33). Colour distinguishes boardings and estimated alightings; line style distinguishes day types.

Bus Cluster Membership: K = 4
 (3383 LSOAs, StopArea night-activity clustering population)

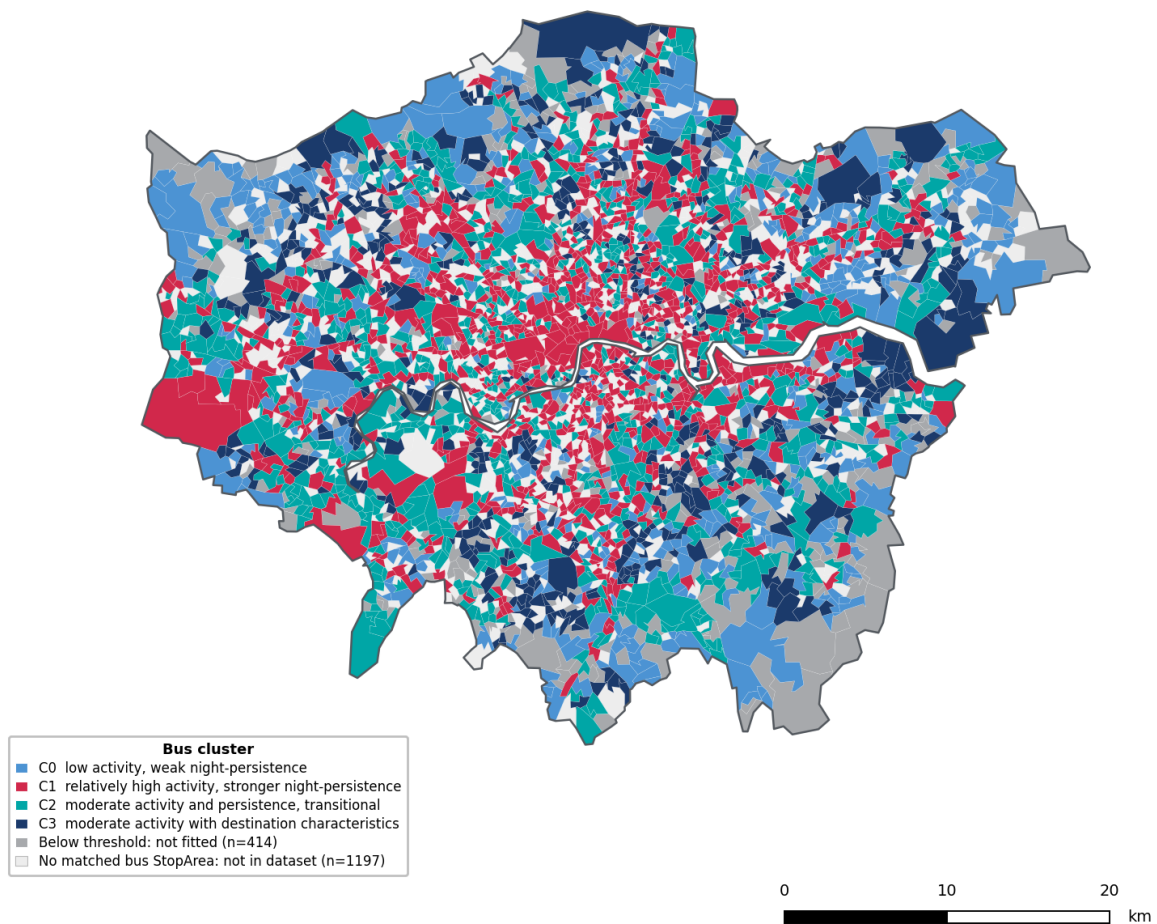


Figure 4.7: Bus cluster map: LSOA unit

Post-clustering behavioural descriptors of the four Bus usage types.

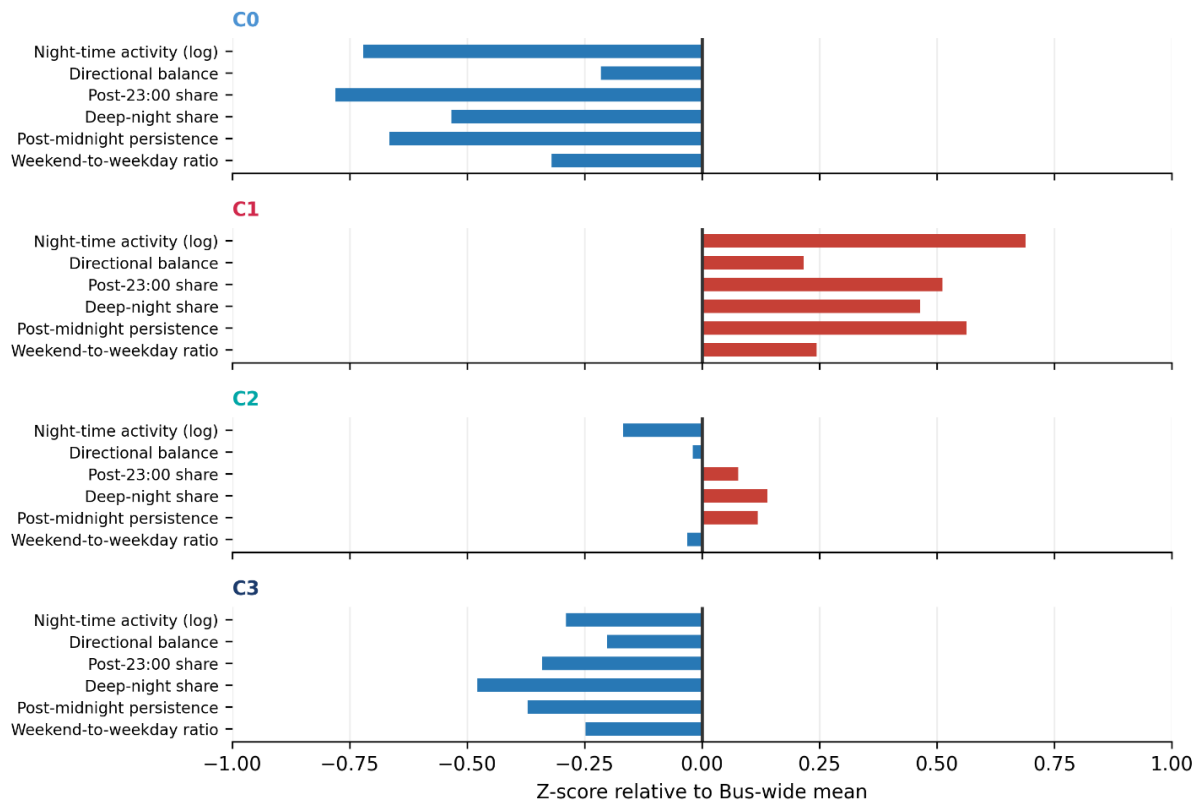


Figure 4.8: Standardised post-clustering behavioural descriptors of the four Bus usage types. Values show cluster-mean z-scores relative to the Bus-wide mean and standard deviation.

C1: Relatively high activity, stronger night-persistence (n = 1,134). C1 contains LSOAs with relatively high levels of night-time Bus activity and the strongest late-night persistence among the four clusters. Activity declines more gradually through the evening and remains comparatively sustained after midnight, with the highest post-midnight persistence value (0.155).

C1 is concentrated mainly in central and inner London, where these persistent patterns form the most continuous areas. The cluster also extends outward along several radial corridors, indicating that sustained late-night Bus activity is not confined to the central core.

C2: Moderate activity and persistence, transitional (n = 1,069). C2 is intermediate between the two endpoints: it retains more late-night activity than C0 but less than C1, and is distributed across inner, middle and some outer areas. Spatially C2 overlaps widely with the other clusters in space. This

distribution supports its interpretation as a transitional profile between the two clearer endpoints rather than as a sharply localised area type.

C3: Moderate activity with alighting-oriented estimated activity (n = 576). C3 shows relatively stronger activity in the earlier part of the night, followed by a clearer decline after 23:00. Within the overlapping middle of the Bus gradient, it differs from C2 through a more alighting-oriented balance of estimated activity. Spatially, C3 occurs more often outside the central core and extends across several outer-London LSOAs.

C0: Low activity, weak night persistence (n = 604). C0 defines the low-activity endpoint. Its profile falls most rapidly from the early evening and records the lowest persistence after midnight (0.031), making it the clearest early-declining bus type.

It is concentrated mainly in outer suburban London, in contrast to the more central and persistent C1 pattern.

Overall Bus structure. The four Bus clusters are differentiated mainly by how long activity persists into the night rather than in entirely different temporal shapes. C1 and C0 mark the clearest endpoints. C1 retains the highest level of activity after midnight (0.155), while C0 declines earliest and has the lowest persistence (0.031); C2 occupies an intermediate position. This variation broadly follows a centre-to-outer London gradient.

C3 adds a second distinction: it is similar to C2 in overall activity level, but shows a more alighting-oriented balance of estimated activity. The difference between them is one of direction rather than persistence. But because Bus alighting is estimated rather than observed, this distinction is interpreted more cautiously than the persistence gradient.

4.3 Relationship with Area Context

The clustering results have shown the night-time usage types, spatial distribution, and cross-system differences for the Rail and Bus systems. The following analysis relates these usage characteristics to LNCW and the wider city background variables, and considers what these associations suggest.

4.3.1 Correspondence with the Geography of Night Work

Figures 4.9 and 4.10 compare each transport type with its own eligible LNWC benchmark (ER=1). Night-work geography is associated with both typologies.

Rail's node rhythms correspond to several distinct night-work settings, while Bus variation aligns with a broader contextual gradient.

4.3.1.1 Rail Clusters and LNWC

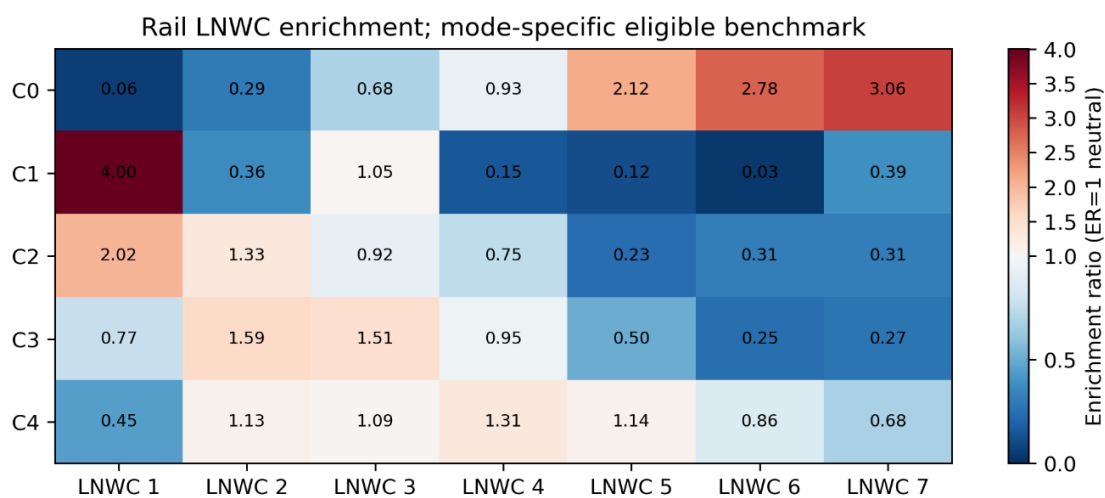


Figure 4.9: Rail LNWC enrichment ratios, using 800 m equal-station catchment context and the 389-station eligible-Rail benchmark (ER=1 neutral).

The seven-part LNWC compositions differed systematically across the five Rail clusters. The observed between-cluster compositional differentiation was greater than expected under random reassignment of the fixed cluster labels ($R^2 = 0.263$, permutation $p = .001$, $n = 389$; 999 permutations). This omnibus result indicates an association between the Rail typology and LNWC composition, while the enrichment ratios below show which LNWC types contributed to the observed differences.

The strongest Rail enrichment links C1 to LNWC 1, Thriving night-worker central hubs (ER = 3.98). The temporal profile establishes that C1 is busy and departure-oriented; the LNWC association places that pattern in the most intensive central night-work setting in London.

C0 sits at the opposite end of the LNWC distribution, with its strongest enrichment in LNWC 7, Night-work periphery (ER = 3.06), alongside LNWC 5 and LNWC 6. These areas all fall within the category of those identified by LNWC as having relatively low levels of night-time activity. This result is consistent with a relatively weak night-time activity context and arrival-oriented profile around these stations.

C4 has a comparatively mixed LNWC profile, consistent with its broad inner-middle-ring spatial distribution rather than one dominant night-work setting.

C3 is most enriched in LNWC 2, Inner suburbs, high night business (ER = 1.59), and LNWC 3, Inner suburbs, low night business (ER = 1.51), with no comparable concentration in LNWC 1. This is notable because C3 was identified earlier as the Rail cluster with the strongest late-night persistence and the highest post-23:00 activity share. This mismatch warrants further examination in the following section.

C2 is also enriched in central and inner-London night-work contexts, including LNWC 1 and LNWC 2, but its contextual profile is more mixed than C1's.

Taken together, LNWC correspondence is strongest at the two ends of the Rail typology. C1 and C0 both exceed an enrichment ratio of 3.0 for their dominant LNWC types, while the association weakens for C2 (2.01), C3 (1.59) and especially C4, which has no dominant type. As these three intermediate clusters contain the majority of analysed stations, LNWC distinguishes only part of the wider Rail typology.

This pattern is consistent with the overall R^2 of 0.263: night-work context is associated with Rail usage type, but does not account for its full variation. The next section examines whether broader urban background characteristics help distinguish the less clearly separated clusters.

4.3.1.2 Bus Clusters and LNWC

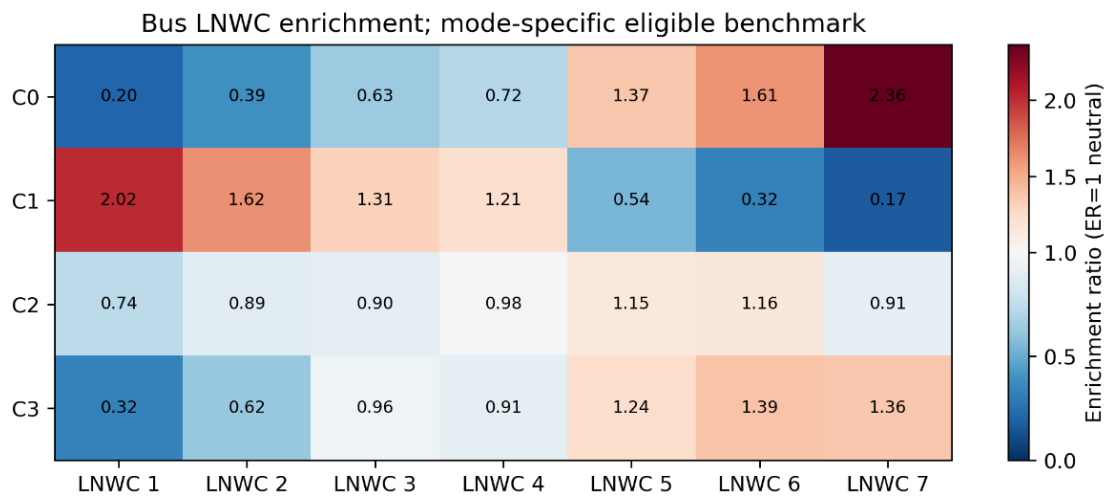


Figure 4.10: Bus LNWC enrichment ratios (ER=1 neutral).

A Pearson chi-square test indicated an association between the four Bus clusters and the seven LNWC types, $\chi^2(18, n = 3,383) = 647.50$, $p < .001$, Cramér's $V = 0.253$, indicating that the association was present but not sufficient to treat Bus cluster membership and LNWC type as equivalent classifications.

C1 and C0 form near mirror images across the LNWC sequence: C1 declines from 2.02 to 0.17 from centre to periphery, while C0 rises from 0.20 to 2.36. C3 shows a weaker peripheral tendency than C0, whereas C2 remains close to neutral throughout (0.74–1.16).

The association again concentrates at the ends of the typology rather than in its middle. In particular, it does little to distinguish C3 from C2 along the directional dimension that separates them in the transport data.

4.3.2 Relation to City Background Variables

The following analysis characterises the context surrounding each cluster. The omnibus tests showed that the contextual distributions differed across clusters for all 20 Bus indicators and for 19 of the 20 Rail indicators after mode-specific Benjamini–Hochberg correction. The exception was the Rail Asian population share (BH-adjusted $p = .054$).

The strongest Rail differentiation concerned car access, younger-adult composition, households with dependent children, private renting and POI intensity.

For Bus, the strongest differences concerned car access, younger-adult composition, POI intensity, private renting and accommodation-and-food employment.

The cluster-versus-rest tests further identified where these overall differences occurred. After Benjamini–Hochberg correction, 63 of the 100 Rail contrasts and 66 of the 80 Bus contrasts remained statistically significant. The z-score panels show the direction of cluster-mean deviation, while rank-biserial correlations quantify the corresponding cluster-versus-rest contrasts. Full omnibus results are reported in Appendix Table A.3, with Rail and Bus cluster-versus-rest effect sizes in Tables A.4 and A.5, respectively.

Mode	Contextual variable	Kruskal–Wallis H	ϵ^2	BH-adjusted p
Rail	No-car household share	165.10	.420	<.001
Rail	Age 20–34 share	144.35	.365	<.001
Rail	Dependent-children share	131.39	.332	<.001
Rail	Private-rented share	105.12	.263	<.001
Rail	Log POI count	102.75	.257	<.001
Bus	No-car household share	735.42	.217	<.001
Bus	Age 20–34 share	583.94	.172	<.001
Bus	Log POI count	518.16	.152	<.001
Bus	Private-rented share	494.95	.146	<.001
Bus	Accommodation-and-food employment share	353.97	.104	<.001

Table 4.1: Strongest contextual differences (top five) across the fixed Rail and Bus clusters.

4.3.2.1 Rail Area Context

Figure 4.14 provides two complementary views of regional background context. The effect-size ranking identifies the dimensions that most clearly differentiate the fixed clusters within each mode; the z-score displays show the direction of each cluster’s profile relative to its mode-wide mean.

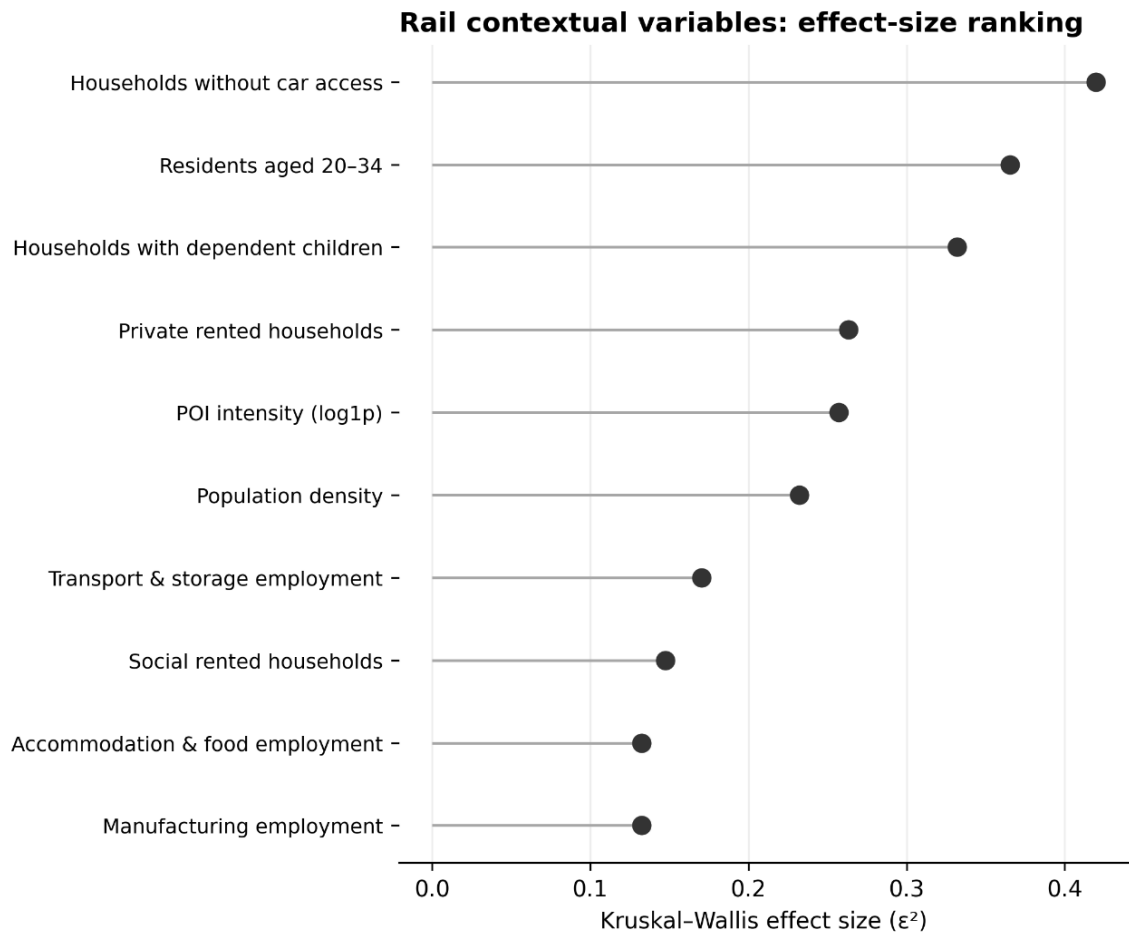


Figure 4.11: Contextual indicators with the largest Kruskal–Wallis effect sizes across the five Rail usage types. Effect sizes are measured using ϵ^2 ; larger values indicate stronger between-cluster differentiation.



Figure 4.12: Standardised contextual profiles of the five Rail usage types. Cell values show cluster-mean z-scores for each contextual indicator relative to all eligible Rail stations. Positive values indicate above-average and negative values below-average contextual characteristics within Rail. Dots indicate significant cluster-versus-rest Mann–Whitney tests after Benjamini–Hochberg correction. The colour scale is mode-specific.

For Rail, vehicle access, age, household composition, tenure and POI intensity provide the strongest overall contextual differentiation. The cluster-level profiles reveal that these dimensions combine into multiple socio-spatial patterns.

Night-time activity centres (C1) are located in younger, more privately rented and amenity-intensive surroundings, with particularly high POI intensity. They also have a higher prevalence of households without cars and fewer households with dependent children, providing a clearer picture of the urban context surrounding C1 stations.

Outer arrival-oriented stations (C0) are associated with a more suburban household profile, including more households with dependent children, fewer car-free households and a smaller share of younger adults. The catchments also have relatively high resident employment in transport and storage, wholesale and retail, health and social work, and manufacturing. This contrasts with the LNWC pattern: C0 lies mainly in peripheral areas with relatively low recorded night-time activity, yet its surrounding residential areas contain comparatively high employment in several industries associated with night-time work.

The inner-middle ring cluster (C4) remains close to the Rail-wide mean across most variables, consistent with its broad spatial coverage.

The central, direction-balanced cluster (C2) retains a more moderate central profile, sharing C1's centrality without its intensity.

Late-persistent stations (C3) return to the mismatch noted earlier. The catchments show higher levels of health deprivation, social renting and unemployment, alongside a larger share of younger adults and residents employed in accommodation and food services.

These characteristics distinguish C3 from both the amenity-intensive central profile of C1 and the more family-oriented outer profile of C0. C3 thus provides a distinctive case for considering how sustained late-night transport use relates to socio-spatial conditions beyond the concentration of night-time work alone.

No-car prevalence requires more cautious interpretation. It is highest around central C1 stations, but these catchments are often highly commercial and contain relatively small residential populations, making

resident-based measures less representative of the wider station environment. For this reason, the observed pattern is better interpreted as a characteristic of the central-area context rather than as a direct indication of passenger characteristics.

4.3.2.2 Bus Area Context

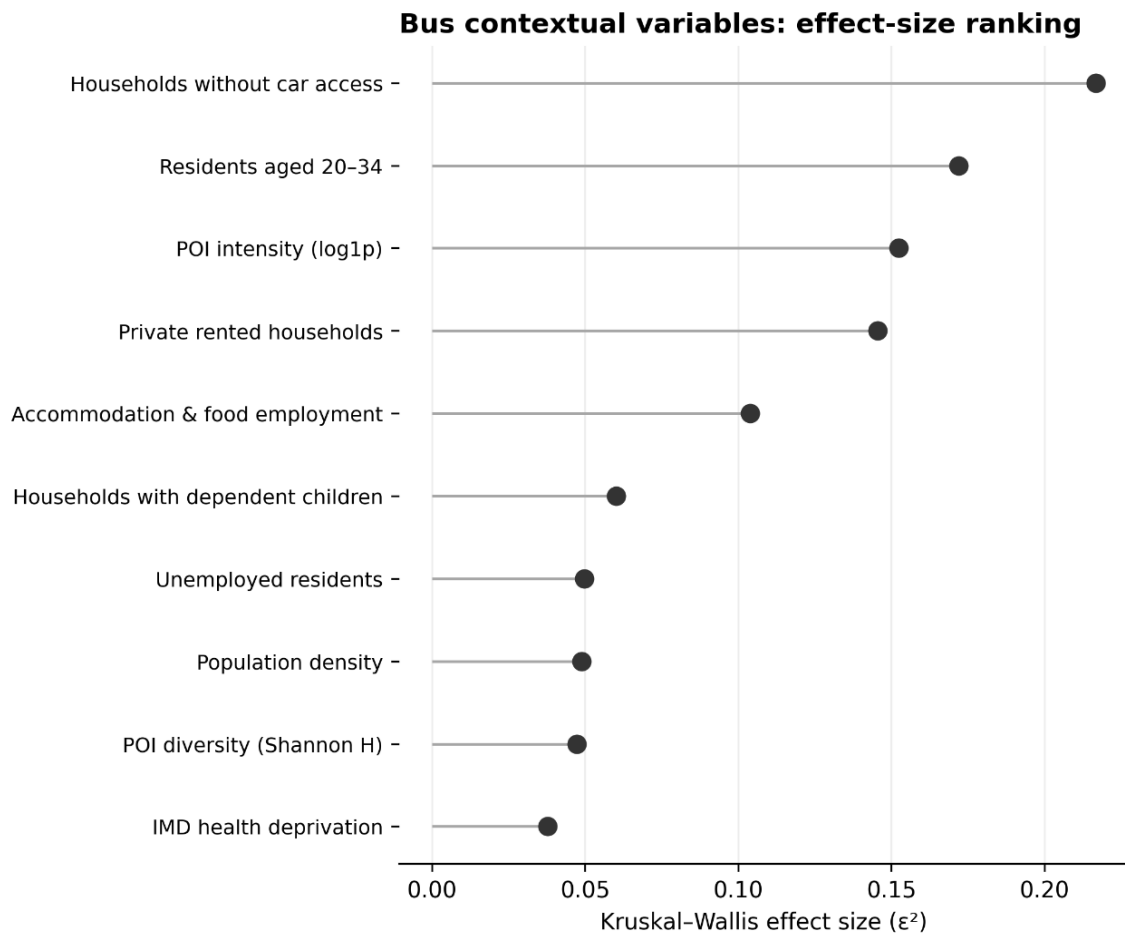


Figure 4.13: Contextual indicators with the largest effect sizes across Bus usage types.

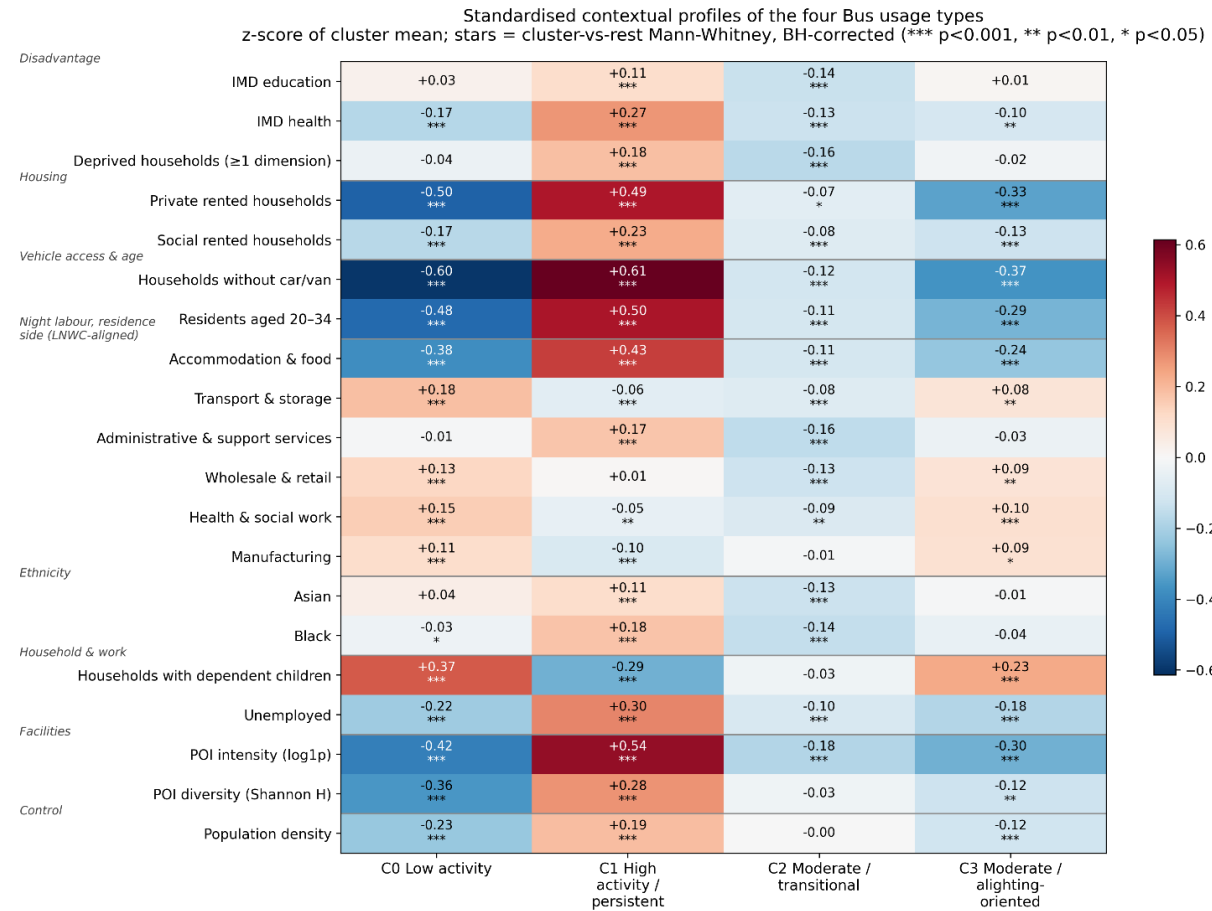


Figure 4.14: Standardised contextual profiles of the four Bus usage types. Cell values show cluster-mean z-scores for each contextual indicator relative to all eligible Bus-served LSOAs. Positive values indicate above-average and negative values below-average contextual characteristics within Bus. Dots indicate significant cluster-versus-rest Mann–Whitney tests after Benjamini–Hochberg correction. The colour scale is mode-specific.

For Bus, car access, younger-adult composition, POI intensity and private renting provide contextual distinctions across the area activity-persistence gradient.

The persistent central/inner C1 cluster is associated with younger populations, more private renting, higher no-car prevalence and greater POI intensity, alongside somewhat higher deprivation-related indicators.

The early-declining outer C0 cluster shows the opposite pattern and a higher share of households with dependent children.

Despite its lower activity and earlier decline, C0 shows above-average residence-based employment in several sectors associated with night-time work, including transport and storage, wholesale and retail, health and social work, and manufacturing, while accommodation and food employment is lower.

C2 and C3 remain intermediate across most of the 20-variable profile, matching their overlapping temporal and spatial positions.

4.4 Summary of Results

The analysis revealed different forms of night-time organisation across the two modes. Rail comprised relatively differentiated types distinguished by direction, spatial position, activity scale and late-night continuation, whereas Bus followed a smoother activity–persistence gradient from central and inner London towards earlier-declining outer areas.

These mode-specific rhythms were also embedded in contrasting socio-spatial contexts. Rail types were associated with distinct combinations of night-work, household, housing, employment and facility characteristics, while Bus context varied more gradually along a centre-to-periphery gradient. These findings provide the basis for the interpretation developed in Chapter 5.

Chapter 5

Discussion

This chapter interprets the empirical findings presented in Chapter 4 in relation to the research questions and the wider literature on night-time mobility. It first considers how the identified night-time usage types are organised across Rail and Bus, before examining their relationships with the London Night Workers Classification and wider socio-spatial context. It then discusses selected transport settings that warrant closer investigation, followed by the implications and limitations of the analysis.

5.1 Overall Organisation of Night-time Public Transport Use

London's night-time public transport use is organised through recurrent and geographically patterned rhythms rather than a single, uniformly declining off-peak condition. Within the common 18:00–05:00 analytical window, locations differ in when activity is concentrated, how directional flows are balanced, how far activity continues into the night, how profiles vary across day types and where these patterns occur in the city. These differences are expressed in different ways across Rail and Bus. The result gives empirical substance to accounts of the urban night as a period with its own uneven temporal and spatial organisation rather than simply a lower-intensity continuation of daytime activity (Schwanen et al., 2012; Van Liempt et al., 2015).

Taken together, the results show that London's night-time public transport use has clear but mode-specific temporal and spatial organisation. Rail and Bus differ in the dimensions through which this organisation is expressed, while the identified transport types also show systematic but incomplete associations with

night-work geography and wider socio-spatial context. The strongest correspondence occurs at the central and peripheral ends of the typologies, whereas intermediate and late-persistent types indicate that transport-use patterns retain temporal and network characteristics not captured by area context alone.

5.2 Different Forms of Night-time Organisation in Rail and Bus

The two modes express night-time organisation through different dominant dimensions. Rail forms a relatively differentiated station typology, whereas Bus varies more continuously along an activity–persistence gradient.

For Rail, the contrast between central departure-oriented and peripheral arrival-oriented stations suggests that London’s centre–periphery structure remains visible after conventional commuting hours, re-expressed through evening dispersal and return flows. This broadly accords with previous station typologies identifying central, residential and other functional roles (Gan et al., 2020; Verma et al., 2021; Zhang et al., 2021). The dedicated night-time window, however, reveals further differentiation. C2 combines relatively balanced entry and exit activity with several major interchange stations, suggesting that network role may also shape directional patterns. Activity intensity and duration likewise diverge: C1 records the highest overall activity but a below-average post-23:00 share, whereas C3 has lower volume but the strongest late-night persistence. These roles also vary across day types, particularly through directional shifts and later continuation on Friday and Saturday nights. This adds a night-time perspective to the intra-week variation identified by Cheng et al. (2024), showing how such variation is expressed through direction and persistence within the dedicated night-time window. Rail night-time use is therefore organised through spatial position, direction, network role, duration and day type rather than centrality alone.

Bus shows a more continuous structure. Its four profiles share a similar evening decline and differ mainly in activity level and persistence, producing a broad transition from more persistent central and inner areas to earlier-declining outer areas. C2 and C3 occupy overlapping intermediate positions, while weekend variation does not alter this overall structure substantially. Bus is therefore better interpreted as a graded activity–persistence pattern than as four equally distinct functional area types. This is consistent with the more distributed nature of Bus activity, where multiple stops and local functions are combined within

each LSOA.

Taken together, Rail concentrates night-time variation into differentiated nodes, while Bus distributes it across a more continuous area-level gradient. Differences in analytical representation may reinforce this contrast, but the two modes nevertheless reveal different dominant forms of night-time organisation.

5.3 Systematic but Incomplete Correspondence with Night-work and Wider Area Context

RQ2 asks how far the transport types are associated with night-work and wider socio-spatial context. The answer is neither that context fully explains the typologies nor that the associations are incidental. Both modes show systematic correspondence with the independently derived London Night Workers Classification (Mavrogeni et al., 2025), but the form of that correspondence differs. Rail clusters are differentiated in their LNWC compositions, whereas Bus clusters show a categorical centre-to-periphery association. Within both modes, the clearest correspondence occurs at the typological endpoints. Central, intensive night-work environments are over-represented around Rail C1 and Bus C1, while peripheral LNWC types are over-represented around Rail C0 and Bus C0. The weaker correspondence among intermediate types therefore shows that night-work geography helps interpret the transport patterns without reproducing them.

The wider contextual indicators clarify what distinguishes these central and peripheral areas. The central types combine intensive observed activity with younger-adult populations, private renting, low household car access and high facility intensity. These areas also combine employment, residential, interchange and amenity-related functions. This extends previous links between station rhythms and surrounding urban functions (Gan et al., 2020) by showing that the central night-time types are also embedded in a distinctive residential and mobility context.

The outer types show the opposite contextual pattern: lower observed activity, earlier contraction, more households with dependent children, greater car access and lower facility intensity. Yet these areas also contain above-average resident employment in several sectors associated with night-time work.

Because these measures describe residents' industries of employment rather than workplace locations,

they do not show that these workers used the observed services. Instead, the coexistence of weak realised transport use and relatively high resident employment in night-related sectors points to a possible separation between the residential geography of night work and the geography of night-time mobility. This provides a concrete transport-based example of the broader distinction between residence, employment and mobility conditions discussed in previous London research (McArthur et al., 2019; Smeds et al., 2020).

Area context is therefore clearly associated with the transport types, but it does not fully explain them. Centrality and night-work intensity account for much of the broad contrast, particularly at the typological endpoints, but not for every temporal role. The incomplete correspondence is itself informative: LNWC captures an important part of the geography of night-time transport, while the transport profiles additionally distinguish mode-specific features such as direction, interchange role and late-night persistence.

5.4 Exceptional Types and Night-time Environments Requiring Closer Attention

Rail C3 provides the clearest example of where transport rhythms cannot be explained by central night-work intensity alone. It has the strongest late-night persistence but is not concentrated in the most facility-intensive central night-work setting. Its inner and eastern catchments instead combine a younger-adult profile with higher health deprivation, unemployment, social renting and accommodation-and-food employment. This combination complicates an account of late-night mobility centred mainly on entertainment districts and high-volume central activity. C3 therefore provides a transport-based example of the broader and more heterogeneous urban night described by Shaw (2014, 2022), in which work, care, maintenance and local services may follow temporal and spatial geographies that do not coincide with those of nightlife.

One plausible reading is that persistent Rail activity partly reflects on-site or less temporally flexible work, an interpretation compatible with the higher accommodation-and-food employment observed around C3. This remains only one possible explanation: interchange, leisure, mixed land use and late-running services may also contribute to the observed persistence. C3 matters because several observations converge on an environment distinct from both the central high-volume and outer early-declining types, rather than

because it establishes a particular passenger purpose.

The outer early-declining Rail and Bus settings deserve attention for a different reason. Weak realised use may coexist with residential concentrations of employment in sectors commonly associated with non-standard hours. Without origin-destination, service-frequency and time-dependent accessibility data, this cannot be labelled unmet demand or inadequate provision. It does, however, define a focused research question: whether early contraction reflects lower underlying travel needs, different modal choices, existing service conditions, dispersed destinations, or some combination of these. These cases are therefore useful for identifying where additional research is needed.

5.5 Implications for Night-time Transport Research and Planning

The findings suggest that night-time public transport should be understood as a temporally differentiated system rather than as a single off-peak condition, consistent with recent work emphasising the temporal heterogeneity of urban accessibility and mobility patterns (Ryan et al., 2023).

The contrast between Rail and Bus further supports a mode-sensitive approach. Rail is organised through differentiated node roles involving direction, station function and persistence, whereas Bus follows a more continuous area-level activity–persistence gradient. Applying a common interpretation across modes would therefore obscure important differences in how night-time transport use is spatially and temporally organised.

More broadly, neither total activity nor spatial centrality alone is sufficient to identify the night-time transport role of a location. In Rail, high-volume central stations are not necessarily the most persistent, showing that intensity and late-night continuation represent distinct planning-relevant dimensions. Together, these findings provide an adaptable framework for examining night-time transport beyond conventional commute-centred and all-day ridership analyses, while also providing a more differentiated basis for planning. By distinguishing temporal and mode-specific transport roles, the typologies can help transport authorities target questions of service continuity, interchange and time-dependent accessibility more precisely across the network.

5.6 Limitations and Future Research

Several limitations define the scope of these interpretations. NUMBAT and BUSTO represent typical day-type profiles rather than continuous observations, so seasonal variation, major events and short-term disruption are not captured. Rail and Bus also differ in spatial unit, temporal resolution and directional measurement, while station catchments and LSOA aggregation simplify the relationship between transport activity and surrounding urban context. Cross-mode comparisons should therefore be interpreted as broad organisational patterns rather than formally equivalent behaviours. The resulting typologies also simplify continuous variation, particularly for Bus where intermediate profiles partly overlap.

A more fundamental limitation is that observed public transport use cannot be equated with underlying transport demand. Smart-card and operational records capture journeys realised under existing service, accessibility and mode-choice conditions. Low observed activity may therefore reflect alternative modes, service constraints or unrealised travel rather than low demand itself. Contextual associations are similarly ecological rather than individual-level or causal: they describe the environments surrounding transport activity but cannot identify who is travelling or why.

Future research could combine these typologies with longer time series, origin–destination or passenger-level data, and measures of service supply and time-dependent accessibility. This would allow analysis to move beyond describing realised night-time mobility towards explaining the mechanisms that produce it and the extent to which observed use reflects underlying demand.

Chapter 6

Conclusion

This dissertation examined how night-time public transport use in London is organised in time and space, how Rail and Bus differ in that organisation, and how the resulting usage types relate to night-work geography and wider socio-spatial context. Separate mode-specific typologies were built for the 18:00–05:00 window. The findings show that night-time public transport use is a temporally differentiated system rather than a single off-peak condition: Rail is organised through direction, station function and persistence around discrete node roles, while Bus follows a more continuous area-level activity–persistence gradient. These usage types are systematically, though not completely, associated with night-work geography, with correspondence strongest at the central and peripheral ends of both typologies. Intermediate and late-persistent types, most notably Rail C3, show temporal and network differences that are not fully captured by the area-level night-work classification and are associated with wider socio-spatial context.

This research makes three contributions. Empirically, it shows that a dedicated night-time window reveals structured temporal and spatial variation that is difficult to distinguish within all-day analysis. Comparatively, it demonstrates that Rail and Bus capture different dimensions of night-time mobility and require mode-sensitive rather than pooled interpretation. Methodologically, it links transport-derived temporal typologies with independently measured socio-spatial context, offering a reproducible framework for comparing mobility-based classifications with urban environmental data without one defining the other.

Practically, the resulting typologies provide a more differentiated basis than total ridership or centrality for

identifying settings where questions of service continuity, interchange and time-dependent accessibility warrant closer attention. These include persistent but not necessarily central Rail nodes and outer areas where early decline coexists with resident employment in night-related sectors. Whether these settings reflect service constraints, mode substitution or lower travel needs remains a question for further research combining these typologies with service-supply, origin–destination and passenger-level data.

Bibliography

- Aitchison, J. (1982). The Statistical Analysis of Compositional Data. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 44(2):139–160.
- Anderson, M. J. (2001). A new method for non-parametric multivariate analysis of variance. *Austral Ecology*, 26(1):32–46.
- Benjamini, Y. and Hochberg, Y. (1995). Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 57(1):289–300.
- Briand, A.-S., Côme, E., El Mahrsi, M. K., and Oukhellou, L. (2016). A mixture model clustering approach for temporal passenger pattern characterization in public transport. *International Journal of Data Science and Analytics*, 1(1):37–50.
- Cao, M. and Hickman, R. (2019). Urban transport and social inequities in neighbourhoods near underground stations in Greater London. *Transportation Planning and Technology*, 42(5):419–441.
- Cats, O. (2024). Identifying human mobility patterns using smart card data. *Transport Reviews*, 44(1):213–243.
- Cheng, Y., Hatzichristos, T., Kostellou, A., Fujiyama, T., Argyropoulou, K., and Spyropoulou, I. (2024). Understanding the intra-day and intra-week ridership patterns of urban rail transit stations in London using a fuzzy clustering approach. *Journal of Public Transportation*, 26:100099.
- Cramér, H. (1999). *Mathematical Methods of Statistics*. Princeton University Press. Available at: <https://books.google.fr/books?id=CRTKKaJO0DYC>.
- Davies, D. L. and Bouldin, D. W. (1979). A Cluster Separation Measure. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, PAMI-1(2):224–227.
- El Mahrsi, M. K., Come, E., Oukhellou, L., and Verleysen, M. (2017). Clustering Smart Card Data for Urban Mobility Analysis. *IEEE Transactions on Intelligent Transportation Systems*, 18(3):712–728.
- Farina, L., Boussauw, K., and Plyushteva, A. (2022). Moving safely at night? Women’s nocturnal mobilities in Recife, Brazil and Brussels, Belgium. *Gender, Place & Culture*, 29(9):1229–1250.
- Fraley, C. and Raftery, A. E. (2002). Model-Based Clustering, Discriminant Analysis, and Density Estimation. *Journal of the American Statistical Association*, 97(458):611–631.
- Gan, Z., Yang, M., Feng, T., and Timmermans, H. (2020). Understanding urban mobility patterns from a spatiotemporal perspective: daily ridership profiles of metro stations. *Transportation*, 47(1):315–336.
- GLA Economics (2024). London at Night: An Updated Evidence Base for a 24 Hour City. Greater London Authority. Available at: <https://data.london.gov.uk/download/24648/5311efaa-47c8-4664-a188-09e8763ae678/London-at-Night-Update-2024%20%28FOR%20PDF%29.pdf>.

- Hadfield, P. (2015). The night-time city. Four modes of exclusion: Reflections on the Urban Studies special collection. *Urban Studies*, 52(3):606–616.
- Hubert, L. and Arabie, P. (1985). Comparing partitions. *Journal of Classification*, 2(1):193–218.
- Kapitza, J. (2022). How people get to work at night. A discrete choice model approach towards the influence of nighttime on the choice of transport mode for commuting to work. *Journal of Transport Geography*, 104:103418.
- Kapitza, J. (2025). Time Matters: Analyzing the Impact of Nighttime on Commuter Trip Chaining Behavior and Travel Time Use in Karlsruhe, Germany. *Transportation Research Record: Journal of the Transportation Research Board*, 2679(2):1242–1256.
- Kim, M.-K., Kim, S.-P., Heo, J., and Sohn, H.-G. (2017). Ridership patterns at subway stations of Seoul capital area and characteristics of station influence area. *KSCE Journal of Civil Engineering*, 21(3):964–975.
- Kim, Y.-L. (2020). Data-driven approach to characterize urban vitality: how spatiotemporal context dynamically defines Seoul's nighttime. *International Journal of Geographical Information Science*, 34(6):1235–1256.
- Kruskal, W. H. and Wallis, W. A. (1952). Use of Ranks in One-Criterion Variance Analysis. *Journal of the American Statistical Association*, 47(260):583–621.
- Lai, J., Cheng, T., and Lansley, G. (2017). Improved targeted outdoor advertising based on geotagged social media data. *Annals of GIS*, 23(4):237–250.
- Lim, S., Ahn, H., Shin, S., Lee, D., and Kim, Y. H. (2024). Investigating night shift workers' commuting patterns using passive mobility data. *Transportation Research Part A: Policy and Practice*, 181:104002.
- Long, Y. and Thill, J.-C. (2015). Combining smart card data and household travel survey to analyze jobs-housing relationships in Beijing. *Computers, Environment and Urban Systems*, 53:19–35.
- Manley, E., Zhong, C., and Batty, M. (2018). Spatiotemporal variation in travel regularity through transit user profiling. *Transportation*, 45(3):703–732.
- Mariñas-Collado, I., Sipols, A. E., Santos-Martín, M. T., and Frutos-Bernal, E. (2022). Clustering and Forecasting Urban Bus Passenger Demand with a Combination of Time Series Models. *Mathematics*, 10(15):2670.
- Martín-Fernández, J.-A., Hron, K., Templ, M., Filzmoser, P., and Palarea-Albaladejo, J. (2015). Bayesian-multiplicative treatment of count zeros in compositional data sets. *Statistical Modelling*, 15(2):134–158.
- Mavrogeni, M., Cheshire, J., Iliev, M., and Van Dijk, J. (2025). Creating the London night-worker geodemographic classification. *Environment and Planning B: Urban Analytics and City Science*, page 23998083251410392.
- McArthur, J., Robin, E., and Smeds, E. (2019). Socio-spatial and temporal dimensions of transport equity for London's night time economy. *Transportation Research Part A: Policy and Practice*, 121:433–443.
- Ministry of Housing, Communities and Local Government (2025). English Indices of Deprivation 2025. Ministry of Housing, Communities and Local Government.

- Office for National Statistics (2021). Census 2021. Office for National Statistics. Available at: <https://www.ons.gov.uk/census>.
- Ordnance Survey (2026). Points of Interest [dataset]. June 2026 release, accessed through EDINA Digimap. Available at: https://digimap.edina.ac.uk/help/our-maps-and-data/os_products/.
- Palm, M., Allen, J., and Farber, S. (2023). Shifted out: The well-being and justice implications of evening and night commuting. *Transportation Research Part D: Transport and Environment*, 122:103875.
- Pan, M., Patwary, A. L., and Uddin, M. (2025). Unlocking nighttime mobility: Land use and accessibility in public transit for night commuters. *Journal of Transport Geography*, 127:104290.
- Peiret-García, C., Kimani, D., and Suel, E. (2026). Beyond the Commute: Temporal Ridership Patterns and Socioeconomic Stratification in London's Underground Stations. Zenodo.
- Plyushteva, A. (2019). Commuting and the urban night: nocturnal mobilities in tourism and hospitality work. *Journal of Policy Research in Tourism, Leisure and Events*, 11(3):407–421.
- Plyushteva, A. and Boussauw, K. (2020). Does night-time public transport contribute to inclusive night mobility? Exploring Sofia's night bus network from a gender perspective. *Transport Policy*, 87:41–50.
- Rousseeuw, P. J. (1987). Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics*, 20:53–65.
- Ryan, J., Pereira, R. H., and Andersson, M. (2023). Accessibility and space-time differences in when and how different groups (choose to) travel. *Journal of Transport Geography*, 111:103665.
- Sari Aslam, N., Ibrahim, M. R., Cheng, T., Chen, H., and Zhang, Y. (2021). ActivityNET: Neural networks to predict public transport trip purposes from individual smart card data and POIs. *Geospatial Information Science*, 24(4):711–721.
- Schwanen, T., Van Aalst, I., Brands, J., and Timan, T. (2012). Rhythms of the Night: Spatiotemporal Inequalities in the Nighttime Economy. *Environment and Planning A: Economy and Space*, 44(9):2064–2085.
- Schwarz, G. (1978). Estimating the Dimension of a Model. *The Annals of Statistics*, 6(2).
- Shaw, R. (2010). Neoliberal Subjectivities and the Development of the Night-Time Economy in British Cities. *Geography Compass*, 4(7):893–903.
- Shaw, R. (2014). Beyond night-time economy: Affective atmospheres of the urban night. *Geoforum*, 51:87–95.
- Shaw, R. (2022). Geographies of night work. *Progress in Human Geography*, 46(5):1149–1164.
- Smeds, E., Robin, E., and McArthur, J. (2020). Night-time mobilities and (in)justice in London: Constructing mobile subjects and the politics of difference in policy-making. *Journal of Transport Geography*, 82:102569.
- Transport for London (2016). Night Tube. Transport for London. Available at: <https://tfl.gov.uk/campaign/tube-improvements/the-future-of-the-tube/night-tube>.

- Transport for London (2024). NUMBAT 2024 [dataset]. Transport for London Open Data. Available at: <https://crowding.data.tfl.gov.uk/> (Accessed 12 August 2026).
- Transport for London (2025). BUSTO 2024/25: Total Demand by Route by Quarter Hour [dataset]. Transport for London Open Data. Available at: <https://crowding.data.tfl.gov.uk/BUSTO/> (Accessed 12 August 2026).
- Van Liempt, I., Van Aalst, I., and Schwanen, T. (2015). Introduction: Geographies of the urban night. *Urban Studies*, 52(3):407–421.
- Verma, T., Sirenko, M., Kornecki, I., Cunningham, S., and Araújo, N. A. (2021). Extracting spatiotemporal commuting patterns from public transit data. *Journal of Urban Mobility*, 1:100004.
- Wang, Q., Zhang, C., Ma, Z., Xi, H., and Yang, H. (2026). Uncovering Travel Rhythms: A Spatiotemporal and Data-Driven Typology of Urban Rail Transit Stations. *Urban Rail Transit*, 12(1):120–144.
- Yang, Y., Samaranayake, S., and Dogan, T. (2023). A clustering-based approach to quantifying socio-demographic impacts on urban mobility patterns. *Environment and Planning B: Urban Analytics and City Science*, 50(9):2452–2469.
- Zhang, Y., Cheng, T., and Aslam, N. (2019). Exploring the Relationship between Travel Pattern and Social-Demographics Using Smart Card Data and Household Survey. *The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, XLII-2/W13:1375–1382.
- Zhang, Y., Marshall, S., Cao, M., Manley, E., and Chen, H. (2021). Discovering the evolution of urban structure using smart card data: The case of London. *Cities*, 112:103157.
- Zhao, Z., Koutsopoulos, H. N., and Zhao, J. (2020). Discovering latent activity patterns from transit smart card data: A spatiotemporal topic model. *Transportation Research Part C: Emerging Technologies*, 116:102627.
- Zhong, C., Batty, M., Manley, E., Wang, J., Wang, Z., Chen, F., and Schmitt, G. (2016). Variability in Regularity: Mining Temporal Mobility Patterns in London, Singapore and Beijing Using Smart-Card Data. *PLOS ONE*, 11(2):e0149222.

Appendix A

Supplementary Figures and Tables

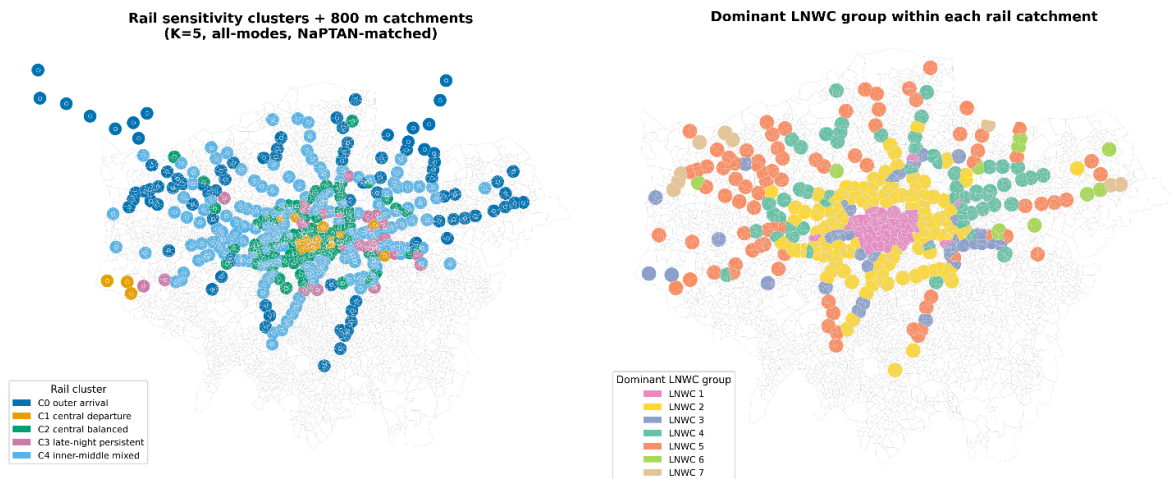


Figure A.1: Supplementary spatial comparison of Rail usage types and LNWC context within the mode-specific contextual-analysis samples

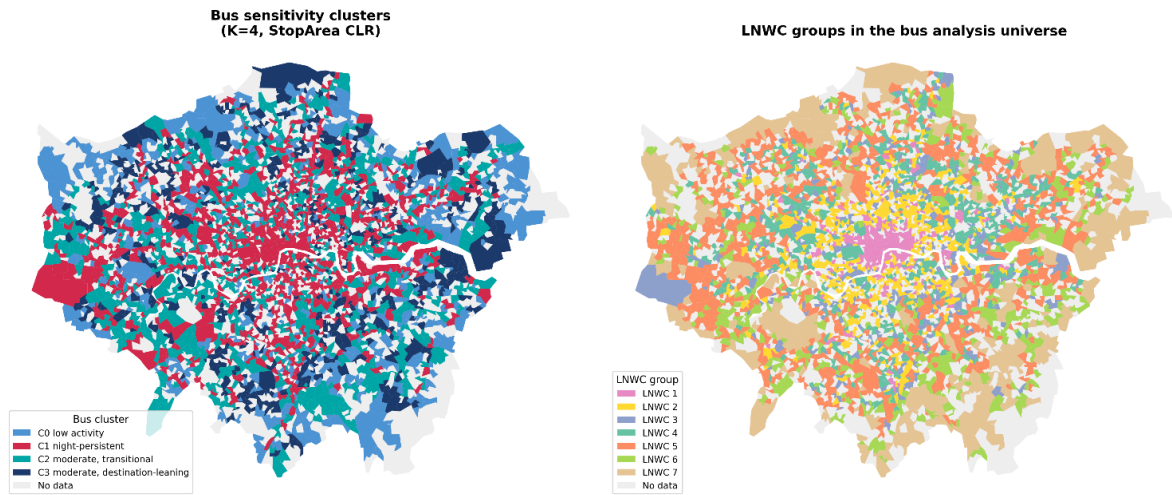


Figure A.2: Supplementary spatial comparison of Bus usage types and LNWc context within the mode-specific contextual-analysis samples

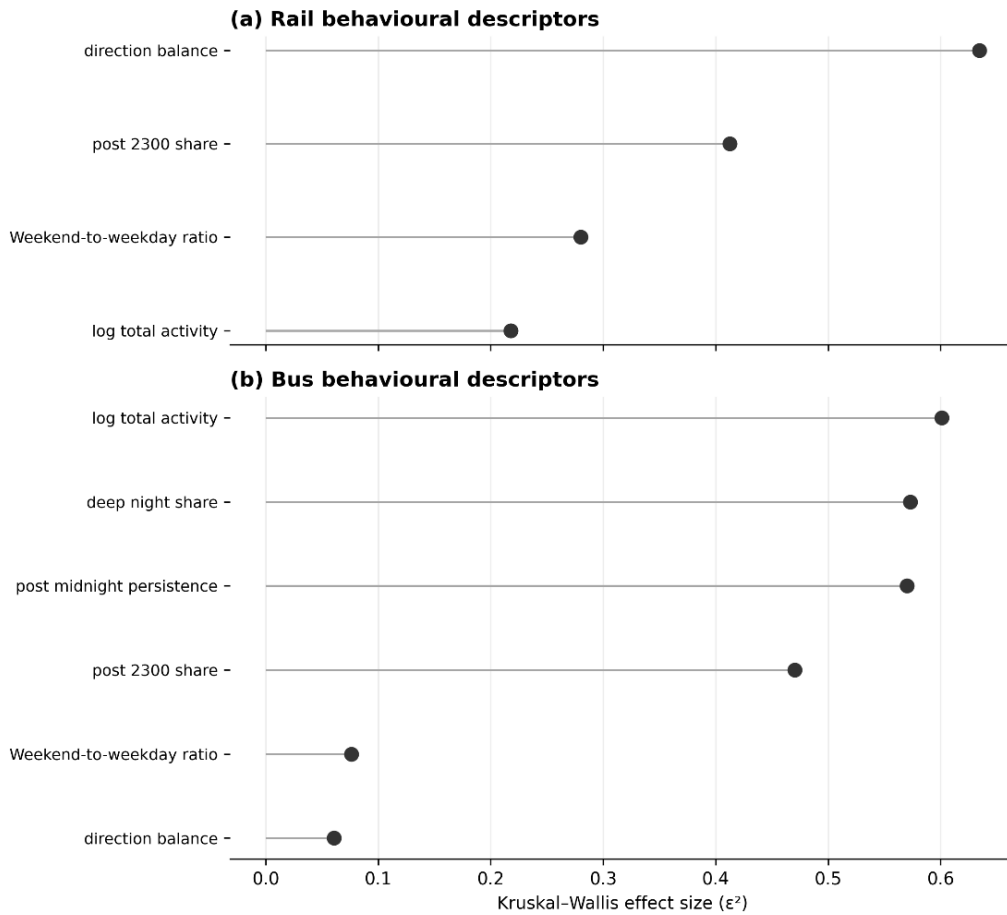


Figure A.3: Effect-size ranking of post-clustering behavioural descriptors across the Rail and Bus usage types: (a) Rail; (b) Bus.

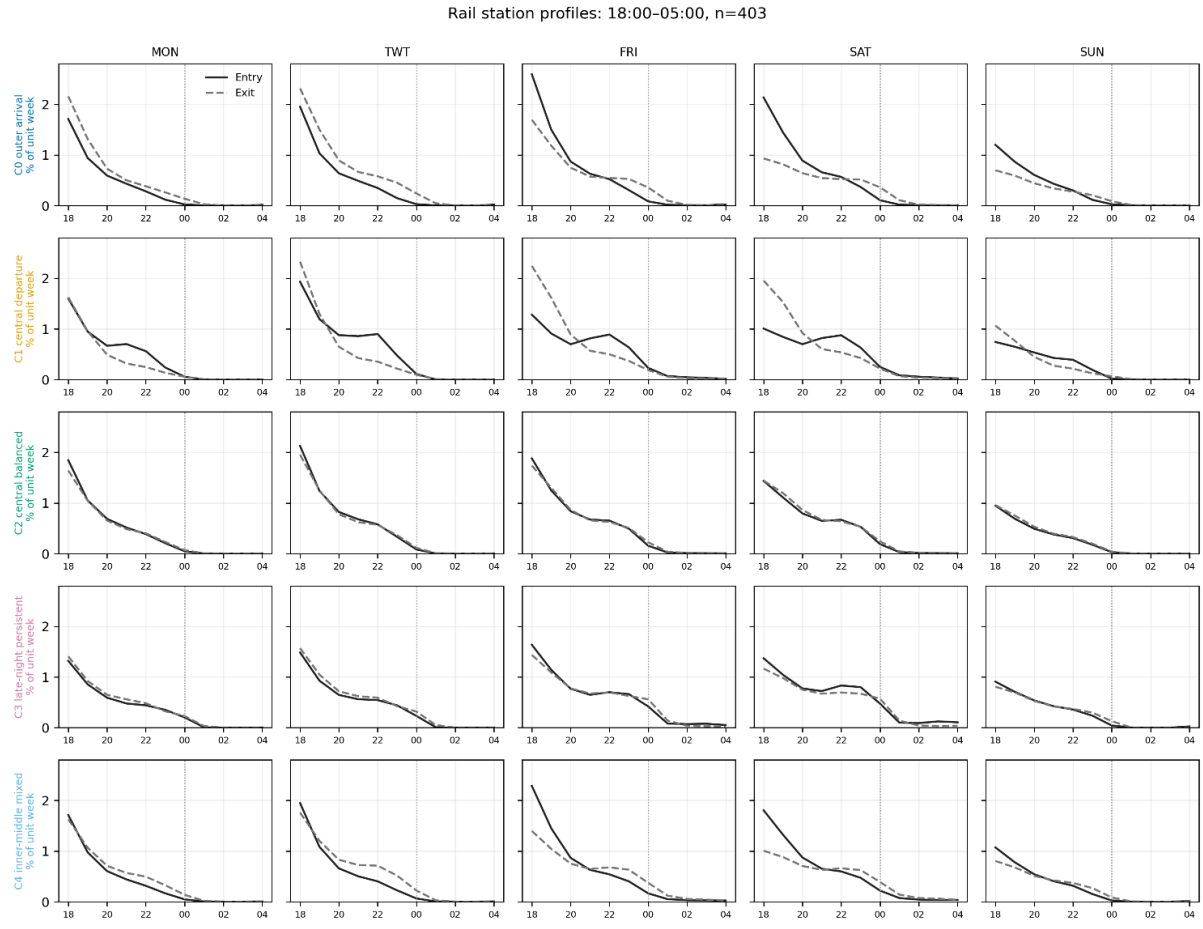


Figure A.4: Original time-series feature map of Rail.

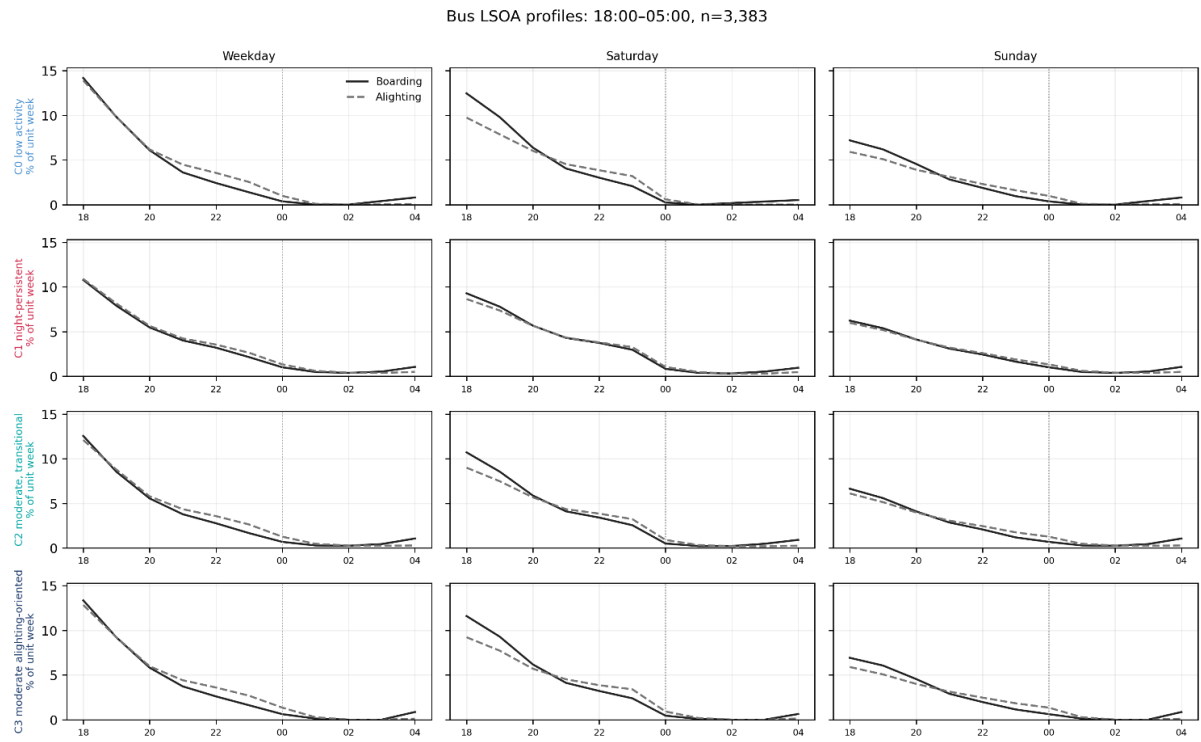


Figure A.5: Original time-series feature map of the Bus.

Table A.1: Mean post-clustering behavioural descriptors of the five Rail usage types

Cluster	n	Log activity	Directional balance	Post-23:00 share	Weekend ratio	Mean km from centre
C0 outer arrival-oriented	89	8.87	-0.52	0.119	0.67	17.8
C1 central departure-oriented	26	10.83	+0.32	0.115	0.85	5.0
C2 central interchange, direction-balanced	90	9.94	+0.02	0.101	0.82	5.8
C3 late-night, extended-duration persistent	31	9.79	-0.13	0.187	0.93	8.4
C4 inner-middle ring mixed	167	9.76	-0.36	0.143	0.81	9.9

Table A.2: Mean post-clustering behavioural descriptors of the four Bus usage types

Cluster	n	Share	Log activity	Directional balance	Post-23:00 share	Persistence	Weekend ratio
C0 low activity, weak night-persistence	604	17.9%	5.57	-0.22	0.084	0.031	0.76
C1 relatively high activity, stronger night-persistence	1,134	33.5%	7.97	-0.09	0.167	0.155	0.83
C2 moderate activity and persistence, transitional	1,069	31.6%	6.40	-0.15	0.140	0.113	0.80
C3 moderate activity with alighting-oriented estimated activity	576	17.0%	6.21	-0.22	0.111	0.061	0.78

Table A.3: Kruskal–Wallis omnibus tests and effect sizes for the 20 contextual indicators by mode

mode	variable	n	k	kruskal_H	p_value	p_bh	epsilon_squared	is_control
bus	no_car_household_ share	3383	4	735.4245	4.37E-159	8.74E-158	0.216758	FALSE
bus	age_20_34_share	3383	4	583.9421	3.05E-126	3.05E-125	0.171927	FALSE
bus	log1p_poi_count	3383	4	518.1648	5.52E-112	3.68E-111	0.152461	FALSE
bus	private_rented_share	3383	4	494.9543	5.92E-107	2.96E-106	0.145592	FALSE
bus	accom_food_share	3383	4	353.974	2.06E-76	8.23E-76	0.103869	FALSE
bus	dependent_children_ share	3383	4	206.4481	1.71E-44	5.68E-44	0.06021	FALSE
bus	unemployed_share	3383	4	170.8384	8.39E-37	2.40E-36	0.049671	FALSE
bus	population_density	3383	4	168.0521	3.35E-36	8.38E-36	0.048846	TRUE
bus	shannon_group	3383	4	163.0298	4.07E-35	9.04E-35	0.04736	FALSE
bus	imd_health	3383	4	130.7925	3.65E-28	7.30E-28	0.03782	FALSE
bus	social_rented_share	3383	4	127.9738	1.48E-27	2.69E-27	0.036985	FALSE
bus	black_share	3383	4	72.91442	1.01E-15	1.69E-15	0.020691	FALSE
bus	deprived_1plus_ share	3383	4	67.71303	1.32E-14	2.03E-14	0.019152	FALSE

(continued on next page)

(Table continued)

mode	variable	n	k	kruskal_H	p_value	p_bh	epsilon_squared	is_control
bus	transport_storage_ share	3383	4	58.4775	1.24E-12	1.78E-12	0.016418	FALSE
bus	admin_support_ share	3383	4	46.04268	5.55E-10	7.41E-10	0.012738	FALSE
bus	imd_education	3383	4	45.40748	7.58E-10	9.47E-10	0.01255	FALSE
bus	manufacturing_share	3383	4	42.59506	3.00E-09	3.53E-09	0.011718	FALSE
bus	health_social_share	3383	4	38.27915	2.47E-08	2.74E-08	0.010441	FALSE
bus	wholesale_retail_ share	3383	4	35.59752	9.11E-08	9.59E-08	0.009647	FALSE
bus	asian_share	3383	4	33.96973	2.01E-07	2.01E-07	0.009165	FALSE
rail	no_car_household_ share	389	5	165.1013	1.18E-34	2.35E-33	0.419535	FALSE
rail	age_20_34_share	389	5	144.3518	3.30E-30	3.30E-29	0.3655	FALSE
rail	dependent_children_ share	389	5	131.3888	1.97E-27	1.31E-26	0.331742	FALSE
rail	private_rented_share	389	5	105.124	7.97E-22	3.99E-21	0.263344	FALSE
rail	log1p_poi_count	389	5	102.7477	2.56E-21	1.02E-20	0.257156	FALSE

(continued on next page)

(Table continued)

mode	variable	n	k	kruskal_H	p_value	p_bh	epsilon_squared	is_control
rail	population_density	389	5	93.20726	2.74E-19	9.14E-19	0.232311	TRUE
rail	transport_storage_ share	389	5	69.43228	2.99E-14	8.55E-14	0.170397	FALSE
rail	social_rented_share	389	5	60.62907	2.14E-12	5.35E-12	0.147472	FALSE
rail	accom_food_share	389	5	54.93032	3.36E-11	7.05E-11	0.132631	FALSE
rail	manufacturing_share	389	5	54.83151	3.52E-11	7.05E-11	0.132374	FALSE
rail	health_social_share	389	5	49.44691	4.71E-10	8.56E-10	0.118351	FALSE
rail	admin_support_ share	389	5	48.11552	8.93E-10	1.49E-09	0.114884	FALSE
rail	imd_health	389	5	46.13406	2.31E-09	3.55E-09	0.109724	FALSE
rail	wholesale_retail_ share	389	5	39.70254	4.99E-08	7.12E-08	0.092975	FALSE
rail	unemployed_share	389	5	39.31275	6.00E-08	8.00E-08	0.09196	FALSE
rail	shannon_group	389	5	31.21757	2.76E-06	3.45E-06	0.070879	FALSE
rail	black_share	389	5	23.4136	0.000105	0.0001231	0.050556	FALSE
rail	imd_education	389	5	14.55205	0.005726	0.0063627	0.027479	FALSE

(continued on next page)

(Table continued)

mode	variable	n	k	kruskal_H	p_value	p_bh	epsilon_squared	is_control
rail	deprived_1plus_ share	389	5	11.32222	0.023172	0.0243913	0.019068	FALSE
rail	asian_share	389	5	9.2858	0.05434	0.0543396	0.013765	FALSE

Table A.4: Cluster-versus-rest rank-biserial correlations for contextual indicators across the five Rail usage types

Variable	C0	C1	C2	C3	C4
imd_education	0.029979	-0.24306	-0.13096	0.305641	0.046394
imd_health	-0.36484	0.08985	0.173244	0.511444	-0.06997
deprived_1plus_share	-0.00229	-0.29625	-0.09	0.193008	0.084426
social_rented_share	-0.4682	0.033482	0.261761	0.469454	-0.04165
no_car_household_share	-0.76195	0.671541	0.490227	0.323842	-0.13972
age_20_34_share	-0.75363	0.570884	0.368413	0.462065	-0.07234
dependent_children_share	0.609342	-0.74041	-0.47217	-0.05695	0.161191
private_rented_share	-0.6597	0.641661	0.288146	0.054244	0.030318
asian_share	-0.0051	0.205552	-0.15273	0.185439	0.006258
black_share	-0.09783	-0.48252	0.019324	0.242026	0.098506
unemployed_share	-0.26395	-0.36279	-0.04586	0.447648	0.159303
accom_food_share	-0.50803	0.20195	0.017094	0.371238	0.147759
transport_storage_share	0.417749	-0.50265	-0.38714	0.110831	0.110428

(continued on next page)

(Table continued)

Variable	C0	C1	C2	C3	C4
admin_support_share	0.176476	-0.57364	-0.28733	0.15246	0.19685
wholesale_retail_share	0.28586	-0.46175	-0.25522	-0.0364	0.132114
health_social_share	0.322123	-0.60097	-0.22007	-0.0164	0.113017
manufacturing_share	0.426157	-0.36427	-0.19926	-0.38836	0.082915
log1p_poi_count	-0.3958	0.79021	0.396024	0.104343	-0.26844
shannon_group	-0.27329	0.088154	0.187663	0.397369	-0.10401
population_density	-0.59278	-0.38885	0.331029	0.388178	0.119059

Cluster labels: C0 outer arrival-oriented; C1 central departure-oriented; C2 central interchange, direction-balanced; C3 late-night, extended-duration persistent; C4 inner-middle ring mixed.

Table A.5: Cluster-versus-rest rank-biserial correlations for contextual indicators across the four Bus usage types

Variable	C0	C1	C2	C3
imd_education	0.01586	0.110656	-0.13295	0.01239
imd_health	-0.12938	0.238931	-0.10516	-0.08168
deprived_1plus_share	-0.03249	0.161371	-0.13415	-0.01558
social_rented_share	-0.1488	0.232457	-0.07534	-0.09693
no_car_household_share	-0.42156	0.53217	-0.09383	-0.25823
age_20_34_share	-0.37473	0.477583	-0.09718	-0.21563
dependent_children_share	0.260105	-0.2425	-0.03602	0.167605
private_rented_share	-0.35352	0.426761	-0.04719	-0.23396
asian_share	0.012142	0.105422	-0.1056	-0.01733
black_share	-0.05711	0.172206	-0.12627	-0.01915
unemployed_share	-0.16628	0.269318	-0.0882	-0.11725
accom_food_share	-0.2734	0.377757	-0.0905	-0.17358
transport_storage_share	0.155932	-0.07675	-0.0843	0.088155
admin_support_share	0.015203	0.11986	-0.12653	-0.01126
wholesale_retail_share	0.100824	-0.01113	-0.10503	0.073577
health_social_share	0.108756	-0.06751	-0.06563	0.094001

(continued on next page)

(Table continued)

Variable	C0	C1	C2	C3
manufacturing_share	0.113451	-0.12058	0.0051	0.064625
log1p_poi_count	-0.30617	0.465166	-0.13842	-0.20413
shannon_group	-0.24394	0.234971	-0.03076	-0.07033
population_density	-0.2382	0.235009	-0.00573	-0.11464

Cluster labels: C0 low activity, weak night-persistence; C1 relatively high activity, stronger night-persistence; C2 moderate activity and persistence, transitional; C3 moderate activity with alighting-oriented.

Appendix B

Research Log and Research Reflection

B.1 Research Log

Table B.1 summarises the formal supervision meetings held throughout the dissertation, the principal topics discussed, and the resulting actions taken in the development of the research.

Table B.1: Research log of dissertation supervision meetings.

Date	Stage	Discussion	Action
12 May 2026	Project Scoping and Data Familiarisation	Reviewed the datasets available for the project, including NUMBAT origin–destination data, BUSTO bus ridership data, and Mikaella Mavrogeni’s London Night Workers Classification (LNWC). Discussed the limits of infrastructure-based ridership data, which records where and when travel occurs but not who is travelling, and considered how mobile-phone-derived flow data might complement it in identifying mismatches between transport provision and demand.	Review the literature on night-time mobility and economic activity; begin drafting a main research question and candidate sub-questions grounded in what the available data can and cannot show.
26 May 2026	Research Question Formulation	Presented an initial main research question and three candidate sub-questions structured around usage clustering, correspondence with LNWC, and service–demand mismatch. Discussed the definition of “night-time” for the analysis, proposing an 18:00–06:00 window with fine (30-minute) temporal bins, and considered whether a shorter or offset window would be more defensible.	Firm up the night-time window definition against precedent in the literature and the LNWC classification; refine the sub-question wording.

(continued on next page)

(Table continued)

Date	Stage	Discussion	Action
9 June 2026	Research Question Refinement	Revised RQ1 to an explicit clustering framing (identifying usage types by area) and RQ2 to a correlation framing linking clusters to LNWC and other socio-economic indicators; RQ3 (service–demand mismatch) retained with minor wording changes. Howard raised a definitional tension: the LNWC classification spans 18:00–05:00, but his substantive interest was in the stricter overnight period (00:00–05:00) least served by daytime transport; discussed whether splitting the clustering by time window would be needed to keep RQ1 and RQ3 compatible.	Retain a single 18:00–05:00 clustering window to preserve the RQ2 linkage to LNWC, while noting a narrower overnight sub-analysis as a possible later refinement.

(continued on next page)

(Table continued)

Date	Stage	Discussion	Action
25 June 2026	Data Construction and Aggregation Strategy	Presented the decision to aggregate bus stops to LSOA level rather than clustering at stop level, because a large share of the roughly 20,000 individual bus stops had sparse or missing counts, particularly at night. Reported that restricting clustering to a “night-peak” sub-window, rather than the full 18:00–05:00 period, produced weaker, less-separated clusters, so the full window was reinstated. Discussed the resulting difference in temporal resolution between rail (15-minute bins, per TfL guidance) and bus (1-hour bins, per BUSTO guidance), and whether combining modes into a single feature vector was feasible given the different vector lengths.	Finalise LSOA-level aggregation for bus; retain the full 18:00–05:00 window; run rail and bus clustering as separate models with mode-specific temporal resolution rather than forcing a shared vector length.

(continued on next page)

(Table continued)

Date	Stage	Discussion	Action
7 July 2026	Feature Engineering Review	Re- Presented an exploratory four-cluster solution built on full-week station/LSOA vectors. In reviewing how entries and exits were represented, the supervisors identified that the feature construction subtracted exits from entries into a single net-flow value per time bin, which discards directional information; clarified that entries and exits should instead be represented as two separate vectors per unit.	Rebuild the feature vectors to preserve entries and exits separately, later formalised as the direction-specific normalisation step (Methodology §3.3.1).

(continued on next page)

(Table continued)

Date	Stage	Discussion	Action
17 July 2026	Scope Review and Methodological Challenge	Reviewed the progress presentation and document. With roughly five weeks to submission, narrowed the dissertation's scope to RQ1 (clustering) and RQ2 (LNWC and socio-spatial linkage); RQ3 (service-demand mismatch) was deprioritised as time-permitting rather than a committed deliverable. Howard questioned the plausibility of the then-current bus K=3 solution, in which one cluster grouped inner-London areas (Westminster, Camden) together with outer-London areas (Kingston, Richmond Park), asking whether this reflected a genuine usage pattern or an artefact of aggregating stop-level data to LSOA level. Clara reframed the partial overlap between the clustering results and Mikaella's LNWC classification as a validation opportunity rather than a concern about novelty.	Tighten and better justify the existing RQ1/RQ2 analysis rather than expanding scope; investigate the aggregation and low-ridership issues behind the implausible bus cluster geography; test whether including full National Rail alongside the Underground changes the rail result.

(continued on next page)

(Table continued)

Date	Stage	Discussion	Action
28 July 2026	Bus Methodology Endorsement	Presented a revised bus clustering pipeline using a minimum-ridership threshold and a centred-log-ratio (CLR) transformation of activity shares, alongside a refitted all-modes rail solution after removing mistakenly included non-London stations. Clara endorsed the CLR-based bus solution as substantively convincing, citing a clean central-London cluster, a mixed-residential ring, an outer ring, and a Heathrow grouping with central London consistent with prior TfL cluster analyses, but did not accept excluding low-ridership stops outright, suggesting instead that clustering jointly on activity shape and raw volume might let low-volume stops separate out naturally. New requests: examine the spatial relationship between bus clusters and rail/Underground stations, interpret the direction of entry/exit imbalance (residential origin versus destination) rather than only its spatial pattern, and add area deprivation (IMD) as an independent variable in RQ2.	Reconcile the shape-versus-volume tension in the bus clustering methodology; add a bus–rail spatial proximity analysis and a direction-of-flow interpretation to the results; add IMD to the RQ2 variable set.

(continued on next page)

(Table continued)

Date	Stage	Discussion	Action
4 August 2026	Analysis Closure and Write-up Priorities	Reviewed submitted draft Methodology and Results chapters. Esra, Clara and Howard agreed that the analysis phase was complete and that no further modelling was required before submission. Howard raised the dissertation's central methodological caution, illustrated with a Leicester Square example: station-level entry/exit counts describe who uses an area, not who lives there or who they are, so results must be framed at the area level rather than attributed to individuals. Esra reinforced this and set the rule that findings should be written as "areas of type X have characteristic Y" rather than "people in area X do Y." Additional feedback covered explaining technical terminology (GMM, BIC, ARI) for a non-specialist reader, moving detailed K-selection justification to the appendix, and ensuring figures were self-explanatory. RQ3 (service-demand mismatch) was confirmed out of scope for the final submission.	Revise the Results and Discussion chapters to consistently use area-level, non-individual-level framing; move detailed K-selection technical justification into the appendix; simplify terminology in the main text; finalise the narrative arc connecting research questions, results and discussion ahead of submission.

B.2 Research Process Reflection

My dissertation did not begin with a fixed research design; it began with a dataset and an open question about what that dataset could support. At the first supervision meeting, before any research questions had been written down, the discussion centred on a basic limitation of infrastructure-based transport data: entry and exit counts show where and when travel happens, but not who is travelling. That distinction, raised by Esra Suel in the very first meeting, turned out to be the single idea that shaped the whole project. Every later version of the research questions, and eventually the ecological-fallacy caution that closed the analysis phase in August, traces back to that early observation.

The early stage of the project was consequently more exploratory than I expected. Between May and early July, the research questions were rewritten several times, not because the underlying interest changed, but because each supervision meeting exposed a gap between what I had proposed and what the data could actually support. The definition of “night-time” is a small but representative example: I first tried a narrower “night-peak” sub-window, expecting it to sharpen the contrast between usage types. Instead, restricting the window weakened the clustering, so I returned to the full 18:00–05:00 window already used by the LNWC classification, a decision that also preserved a clean way to connect the clustering results to LNWC in RQ2. Rather than presupposing the right time frame in advance, I had to let the clustering evidence and the analytical fit tell me which one worked.

A second recurring theme was that data-preprocessing decisions I initially treated as minor turned out to have first-order consequences. The bus data illustrates this most clearly. Aggregating roughly 20,000 individual stops up to LSOA level was necessary because most stops had sparse or missing counts at night, but that aggregation then interacted with the shape of the resulting clusters in ways that took most of July and August to fully understand: raw counts dominated the cluster structure more than the shape of usage over time, one solution mixed geographically implausible areas together, and a subsequent compositional (CLR) transformation intended to fix this introduced its own distortion around stops with many zero-activity bins. Reaching a bus clustering solution that was both statistically defensible and substantively plausible to supervisors

who know the network required testing several intermediate fixes in sequence rather than arriving at the answer directly. The equivalent process for rail was comparatively short, largely because the Underground network has far less low-ridership noise to contend with.

Supervision meetings consistently functioned as a corrective against scope growth rather than a source of new directions. The clearest example was the 17 July meeting, where, with roughly five weeks left, the three-question design built around clustering, LNWC correspondence and service–demand mismatch was narrowed to the first two; the third question was not abandoned so much as postponed indefinitely by time pressure. I found this useful discipline: it forced me to make the existing clustering and LNWC analysis more defensible rather than adding a new analytical strand to compensate for unresolved questions in the first two.

The final and most consequential piece of guidance came on 4 August, after the analysis itself was declared finished. Howard’s Leicester Square example, that a station’s night-time users are not the same people as the area’s night-time residents, gave precise language to the distinction Esra had raised in the first meeting. It meant that even a statistically solid result had to be rewritten carefully: claims had to describe area-level types and their characteristics, never the people using them. Managing that boundary consistently across the Results and Discussion chapters, while still making the analysis feel substantively meaningful, was the last, and in retrospect the most important, methodological skill the dissertation required